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PURE MATHEMATICS WITH STATISTICS • UPPER SIXTH

UPPER SIXTH — PURE MATHEMATICS WITH STATISTICS

CHAPTER 11

RANDOM VARIABLES AND DISCRETE RANDOM VARIABLES

SUGGESTED TIME: 9 PERIODS

IN THIS CHAPTER

This chapter introduces random variables as numerical summaries of random experiments and develops the full toolkit for discrete random variables: probability distributions, expectation, variance, cumulative distribution functions, mode, median, sums of independent variables and probability generating functions. It provides the essential foundation for the study of standard distributions (binomial, Poisson, normal) later in the course.

- By the end of this chapter I can define a random variable and distinguish between discrete and continuous random variables.
- By the end of this chapter I can construct the probability distribution (table of values and probabilities) of a discrete random variable from a described experiment.
- By the end of this chapter I can verify and use the conditions for a probability mass function, including finding unknown constants.
- By the end of this chapter I can compute the expectation E(X), the variance Var(X) and the standard deviation of a discrete random variable, and use the linearity rules E(aX+b) and Var(aX+b).
- By the end of this chapter I can build and interpret the cumulative distribution function F(x) and use it to compute probabilities.
- By the end of this chapter I can determine the mode and the median of a discrete probability distribution.

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EXERCISES exam style + solutions

SUMMARY SHEET the essentials to remember

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2. Probability distributions and the probability mass function
3. Expectation, variance and standard deviation
4. The cumulative distribution function, mode and median
5. The distribution of $X_1 + X_2$
6. Probability generating functions
7. Integration activity
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CHAPTER OBJECTIVES

- By the end of this chapter I can define a random variable and distinguish between discrete and continuous random variables.
- By the end of this chapter I can construct the probability distribution (table of values and probabilities) of a discrete random variable from a described experiment.
- By the end of this chapter I can verify and use the conditions for a probability mass function, including finding unknown constants.
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- By the end of this chapter I can build and interpret the cumulative distribution function F(x) and use it to compute probabilities.
- By the end of this chapter I can determine the mode and the median of a discrete probability distribution.
- By the end of this chapter I can find the distribution of the sum of two independent discrete random variables.
- By the end of this chapter I can define and use a probability generating function to recover probabilities, expectation and variance.

PREREQUISITES

- Elementary probability: sample spaces, events, P(A), complementary and combined events (Lower Sixth).
- Conditional probability and independence of events, $P(A \cap B) = P(A)P(B)$ for independent events.
- Counting techniques: arrangements, permutations and combinations.
- Summation notation Σ and manipulation of finite sums.
- Basic differentiation of polynomials and power series (useful for probability generating functions).

1 RANDOM VARIABLES AND DISCRETE RANDOM VARIABLES

At the Mokolo market in Yaoundé, a trader sells phone credit cards and never knows in advance how many customers will come each day: some days 3, other days 15. Yet, to avoid losses, she must decide every morning how many cards to stock. Stock too few, and she loses sales; stock too many, and her money sleeps on the shelf. How can mathematics help her make a good decision when the outcome is uncertain? The answer lies in modelling the number of customers by a **random variable**, summarising its behaviour by a **probability distribution**, and computing its **expectation**. This chapter gives you the tools that banks, insurers and even the GCE examiners use to reason about uncertainty.

1.1 Random experiments revisited

Sample space. Recall from your study of probability that a **random experiment** is a process whose outcome cannot be predicted with certainty, even though we know the list of all possible outcomes. The set of all possible outcomes is called the **sample space**, denoted Ω. For example:

- Tossing one coin: $\Omega = \{H, T\}$.
- Tossing two coins: $\Omega = \{HH, HT, TH, TT\}$.
- Rolling a fair die: $\Omega = \{1, 2, 3, 4, 5, 6\}$.
- Counting the trader's daily customers: $\Omega = \{0, 1, 2, 3, \ldots\}$.

The need to attach numbers to outcomes. Outcomes are not always numbers: HH, HT are pairs of letters, not quantities. Yet to compute averages, to compare experiments, to build models for the trader, we need numbers. We therefore invent a rule that attaches a real number to every outcome. For the two coins, a natural rule is: "count the number of heads". Then

$$HH \mapsto 2, \qquad HT \mapsto 1, \qquad TH \mapsto 1, \qquad TT \mapsto 0.$$

This rule is exactly what we call a random variable.

1.2 Definition of a random variable

DEFINITION **Random variable**

A **random variable** X is a real-valued function defined on the sample space Ω of a random experiment:

$$X : \Omega \longrightarrow \mathbb{R}, \qquad \omega \longmapsto X(\omega).$$

To each outcome ω of the experiment, X associates exactly one real number $X(\omega)$.

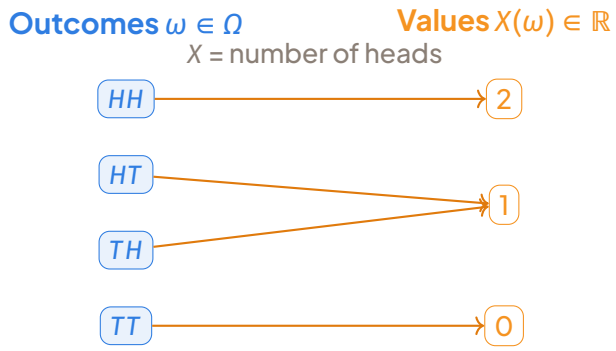
WARNING **A random variable is neither random nor a variable**

The name is misleading. A random variable is not random: the rule X itself is perfectly fixed. It is not a variable in the algebraic sense (like the x in $2x + 1 = 5$): it is a function. The randomness lies in the experiment, not in X. Also, do not confuse X (the function, written with a capital letter) with x (one particular value it can take, written with a small letter). Examiners penalise this confusion.

The event $\{X = x\}$. Since X is a function on Ω, the statement "X takes the value x" describes a set of outcomes, hence an event:

$$\{X = x\} = \{\omega \in \Omega : X(\omega) = x\}.$$

Its probability is written $P(X = x)$. For the two coins, $\{X = 1\} = \{HT, TH\}$, so $P(X = 1) = \frac{2}{4} = \frac{1}{2}$.



The random variable X "number of heads" maps each outcome of tossing two coins to a real number. Note that two outcomes, HT and TH, share the same image 1.

1.3 Discrete and continuous random variables

DEFINITION **Discrete random variable**

A random variable X is **discrete** if the set of values it can take is finite or countably infinite (i.e. its values can be listed as $x_1, x_2, x_3, \ldots$).

By contrast, a **continuous** random variable can take any value in an interval of $\mathbb{R}$ (its set of values is uncountable). Continuous variables will be studied later; this chapter deals only with the discrete case.

EXEMPLE **Discrete or continuous?**

- The number of goals scored by the Indomitable Lions in a match: values 0, 1, 2, … — discrete.
- The number of customers of the Mokolo trader in a day: values 0, 1, 2, … — discrete.
- The score on a fair die: values {1, 2, 3, 4, 5, 6} (finite) — discrete.
- The height of a randomly chosen Upper Sixth student: any value in an interval, say [1.40 m, 2.00 m] — continuous.
- The waiting time for a taxi at the Ngousso junction: any non-negative real number — continuous.

KEY POINTS **Finite versus countably infinite**

- Finite: the die score (6 values), the number of heads in two tosses (3 values).
- Countably infinite: the number of tosses of a coin until the first head appears: 1, 2, 3, … with no upper bound — the list never ends, but it can still be enumerated.

1.4 Probability distribution and probability mass function

DEFINITION **Probability distribution of a discrete random variable**

Let X be a discrete random variable taking values $x_1, x_2, \ldots$ (finite or countably infinite). The **probability distribution** of X is the complete list of pairs (x_i, p_i) where

$$p_i = P(X = x_i).$$

It is usually presented as a table:

x_i	x_1	x_2	$\cdots$	x_n
$P(X = x_i)$	p_1	p_2	$\cdots$	p_n

Since the events $\{X = x_i\}$ are mutually exclusive and exhaustive, we always have

$$0 \le p_i \le 1 \quad \text{for every } i, \qquad \text{and} \qquad \sum_i p_i = 1.$$

DEFINITION **Probability mass function (PMF)**

The function $p_X(x) = P(X = x)$ defined for all real x (with $p_X(x) = 0$ if x is not a possible value of X) is called the **probability mass function** of X. For a discrete random variable, the PMF completely characterises the distribution.

KEY POINTS **Support of a discrete random variable**

The set of values x for which $P(X = x) > 0$ is called the **support** of X. For a discrete variable, the support is exactly the set of values that X can actually take.

METHOD **Building the distribution of X**

1. List all the outcomes of Ω and their probabilities.
2. Compute the value $X(\omega)$ for each outcome.
3. Group the outcomes giving the same value x_i: they form the event $\{X = x_i\}$.
4. Add the probabilities of the outcomes inside each group to get $p_i = P(X = x_i)$.
5. Check that $\sum p_i = 1$. If not, an outcome has been forgotten or counted twice.

WORKED EXAMPLE 1 **Number of heads in two tosses of a fair coin**

Two fair coins are tossed. Let X be the number of heads obtained. Determine the probability distribution of X.

SOLUTION

Solution. The sample space is $\Omega = \{HH, HT, TH, TT\}$, each outcome having probability $\frac{1}{4}$ since the coins are fair and independent.
Applying the method: $X(HH) = 2, X(HT) = 1, X(TH) = 1, X(TT) = 0$. Grouping:

$$P(X = 0) = P(\{TT\}) = \frac{1}{4}, \quad P(X = 1) = P(\{HT, TH\}) = \frac{2}{4} = \frac{1}{2}, \quad P(X = 2) = P(\{HH\}) = \frac{1}{4}.$$

x_i	0	1	2
$P(X = x_i)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

Check: $\frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1$. ✓

WORKED EXAMPLE 2 **Score on a fair die**

A fair die is rolled and Y is the score shown. Give the distribution of Y.

SOLUTION

Solution. Here each outcome of $\Omega = \{1, 2, 3, 4, 5, 6\}$ is already a number, so $Y(\omega) = \omega$. Each face has probability $\frac{1}{6}$:

y_i	1	2	3	4	5	6
$P(Y = y_i)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

This is the uniform distribution on $\{1, \dots, 6\}$: all values are equally likely.

WORKED EXAMPLE 3 **Defective items in a sample**

A box contains 5 phone chargers, of which 2 are defective. A quality controller picks 2 chargers at random, without replacement. Let Z be the number of defective chargers in the sample. Find the distribution of Z.

SOLUTION

Solution. The possible values of Z are 0, 1, 2. The number of ways to choose 2 chargers out of 5 is $\binom{5}{2} = 10$, all equally likely.

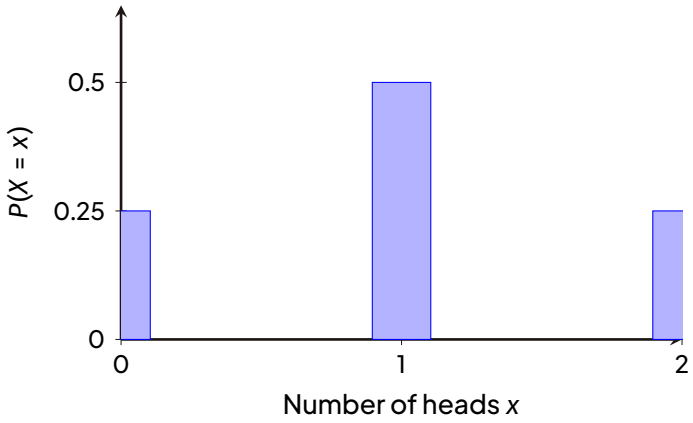
$$P(Z = 0) = \frac{\binom{2}{0}\binom{3}{2}}{10} = \frac{3}{10},$$
$$P(Z = 1) = \frac{\binom{2}{1}\binom{3}{1}}{10} = \frac{6}{10},$$
$$P(Z = 2) = \frac{\binom{2}{2}\binom{3}{0}}{10} = \frac{1}{10}.$$

z_i	0	1	2
$P(Z = z_i)$	$\frac{3}{10}$	$\frac{6}{10}$	$\frac{1}{10}$

Check: $\frac{3}{10} + \frac{6}{10} + \frac{1}{10} = 1$. ✓

1.5 Representing a distribution by a bar chart

A probability distribution is conveniently pictured by a **bar chart**: the values x_i are placed on the horizontal axis and the heights of the bars equal the probabilities p_i. The total of the heights is always 1. (For a discrete variable, a histogram with bars of equal width is also acceptable, but the bar chart emphasises the gaps between possible values.)



Bar chart of the probability distribution of X, the number of heads when two fair coins are tossed. The symmetry reflects the fairness of the coins.

WARNING **Common mistakes**

- Forgetting to add probabilities when several outcomes give the same value of X (e.g. writing $P(X = 1) = \frac{1}{4}$ instead of $\frac{1}{2}$ for the two coins).
- Giving a table whose probabilities do not sum to 1 — always check!
- Confusing $P(X = x)$ with $P(X \leq x)$: the latter is the cumulative probability, studied later in this chapter.
- Believing a discrete variable can take "any" value: between two consecutive values of a discrete variable, no value is possible (a die never shows 3.5).
- Writing $P(X = x)$ for a value x outside the support; remember $P(X = x) = 0$ for such x.

DID YOU KNOW? **Why insurers love random variables**

When an insurance company in Douala fixes the price of a policy, it models the number of accidents a driver will have in a year as a discrete random variable, estimates its distribution from past data, and computes its expectation. The premium must exceed this expected cost, otherwise the company loses money on average. The same mathematics powers mobile-money risk management and the design of fair games of chance.

EXERCISE 1 **Practice**

1. Three fair coins are tossed and X counts the number of heads. Construct the probability distribution of X and draw its bar chart.
2. A bag contains 4 red and 6 black balls. Two balls are drawn at random without replacement, and Y counts the red balls drawn. Find the distribution of Y.
3. State, with a reason, whether each variable is discrete or continuous: (a) the number of students absent in your class tomorrow; (b) the rainfall in Bamenda next Saturday, in millimetres.

SOLUTION **Exercise 1**

1. Ω has 8 equally likely outcomes. Grouping by number of heads: $P(X=0)=\frac{1}{8}$, $P(X=1)=\frac{3}{8}$, $P(X=2)=\frac{3}{8}$, $P(X=3)=\frac{1}{8}$. Sum $=1$. ✓ The bar chart has bars of heights $\frac{1}{8},\frac{3}{8},\frac{3}{8},\frac{1}{8}$ at $x=0,1,2,3$.

2. $\binom{10}{2}=45$ equally likely pairs. $P(Y=0)=\frac{\binom{6}{2}}{45}=\frac{15}{45}=\frac{1}{3}$; $P(Y=1)=\frac{\binom{4}{1}\binom{6}{1}}{45}=\frac{24}{45}=\frac{8}{15}$; $P(Y=2)=\frac{\binom{4}{2}}{45}=\frac{6}{45}=\frac{2}{15}$. Sum $=1$. ✓

3. (a) Discrete: the values are whole numbers $0,1,2,\ldots$. (b) Continuous: rainfall can take any non-negative real value within an interval.

2 PROBABILITY DISTRIBUTIONS AND THE PROBABILITY MASS FUNCTION

In the previous section we met the idea of a **random variable**: a numerical quantity X whose value depends on the outcome of a random experiment, and we saw that a **discrete random variable** takes only isolated values, which we can list as $x_1, x_2, x_3, \ldots$ Knowing the possible values of X, however, is only half of the story. A fair die and a loaded die both take the values $1,2,3,4,5,6$; what distinguishes them is how likely each value is. The complete description of a discrete random variable is therefore a list of its values together with the probability attached to each value. That complete description is what we now call a **probability distribution**, and the function that delivers the probabilities is the **probability mass function**. Mastering this object is essential: nearly every GCE question on discrete random variables begins by giving you a distribution and asking you to work with it.

2.1 The probability mass function

DEFINITION **Probability mass function**

Let X be a discrete random variable taking values $x_1, x_2, \dots, x_n$ (or an infinite list $x_1, x_2, \dots$). The **probability mass function** (p.m.f.) of X is the function p defined by

$$p(x) = P(X = x),$$

that is, $p(x_i)$ is the probability that X takes the value x_i. We also write $p_i = P(X = x_i)$. For any x which is not a possible value of X, $p(x) = 0$.

The word mass is suggestive: imagine a total mass of 1 kilogram of probability, chopped into lumps and placed at the points $x_1, x_2, \dots$ on a number line. The lump at x_i has mass p_i. This picture immediately tells us what conditions a genuine p.m.f. must satisfy.

PROPERTY **Defining conditions of a p.m.f.**

A function p is the probability mass function of a discrete random variable if and only if

1. $p(x_i) \geq 0$ for every i (in fact $0 \leq p(x_i) \leq 1$);
2. $\sum_i p(x_i) = 1$, the sum being taken over all possible values of X.

Justification. Condition 1 holds because every probability lies between 0 and 1. For condition 2, the events $\{X = x_1\}, \{X = x_2\}, \dots$ are mutually exclusive (X cannot take two values at once) and exhaustive (X must take some value). Hence, by the addition law for mutually exclusive events,

$$\sum_i p(x_i) = P(X = x_1) + P(X = x_2) + \cdots = P(X \text{ takes one of its values}) = 1.$$

This simple argument is worth being able to reproduce: it is the reason behind the single most used technique in this topic.

KEY POINTS **Presenting a distribution**

A probability distribution is usually presented as a complete table:

x	x_1	x_2	x_3	$\cdots$	x_n
$P(X = x)$	p_1	p_2	p_3	$\cdots$	p_n

The table must show every possible value of X and the probabilities must sum to 1. Always check this sum before using a distribution.

DID YOU KNOW? **Terminology note**

The syllabus mentions “probability density or mass function”. For discrete random vari-

ables the correct term is probability mass function (p.m.f.). The term "probability density function" (p.d.f.) is reserved for continuous random variables, which you will meet later. In the GCE examination, always write "probability mass function" or "p.m.f." when dealing with discrete variables.

2.2 Finding an unknown constant: the classic GCE question

Very often the GCE gives a distribution containing an unknown constant k and asks you to find it. The key is always the same: the probabilities must add up to 1.

METHOD **Finding an unknown constant in a distribution**

1. Write down the equation $\sum_i p(x_i) = 1$ using the given formula or table.
2. Solve the resulting equation for k.
3. Check that with this value of k, every probability satisfies $0 \le p(x_i) \le 1$. (If an equation gives two solutions, one of them usually fails this check and must be rejected.)
4. Rewrite the distribution as a complete numerical table.

WORKED EXAMPLE 4 **A distribution given by a formula**

A discrete random variable X has probability mass function

$$P(X = x) = kx, \qquad x = 1, 2, 3, 4.$$

Find the value of k and write out the probability distribution of X in a table.

SOLUTION

Solution. The probabilities must sum to 1:

$$\sum_{x=1}^{4} kx = k(1) + k(2) + k(3) + k(4) = 10k.$$

Setting $10k = 1$ gives $k = \frac{1}{10} = 0.1$.
Check: the probabilities are 0.1, 0.2, 0.3, 0.4, all between 0 and 1. ✓
The distribution is:

x	1	2	3	4
$P(X = x)$	0.1	0.2	0.3	0.4

WORKED EXAMPLE 5 **A quadratic equation for k**

A random variable X takes the values 0, 1, 2 with

$$P(X = 0) = k, \qquad P(X = 1) = 2k, \qquad P(X = 2) = 4k^2 + k.$$

Find the value of k.

SOLUTION

Solution. Summing the probabilities:

$$k + 2k + 4k^2 + k = 4k^2 + 4k = 1 \quad \Longrightarrow \quad 4k^2 + 4k - 1 = 0.$$

By the quadratic formula,

$$k = \frac{-4 \pm \sqrt{16 + 16}}{8} = \frac{-4 \pm \sqrt{32}}{8} = \frac{-1 \pm \sqrt{2}}{2}.$$

Now $\frac{-1 - \sqrt{2}}{2} \approx -1.207 < 0$, which would make $P(X = 0)$ negative: rejected. Hence

$$k = \frac{\sqrt{2} - 1}{2} \approx 0.207.$$

Check: $P(X = 0) \approx 0.207$, $P(X = 1) \approx 0.414$, $P(X = 2) \approx 4(0.0429) + 0.207 \approx 0.379$; all in $[0, 1]$ and $0.207 + 0.414 + 0.379 = 1$. ✓

WORKED EXAMPLE 6 **An infinite distribution: geometric-type**

A discrete random variable X takes values 1, 2, 3, ... with

$$P(X = x) = k\left(\frac{2}{3}\right)^x.$$

Find the value of k.

SOLUTION

Solution. The sum of probabilities is an infinite geometric series:

$$\sum_{x=1}^{\infty} k\left(\frac{2}{3}\right)^x = k \cdot \frac{\frac{2}{3}}{1 - \frac{2}{3}} = k \cdot \frac{2/3}{1/3} = 2k.$$

Setting $2k = 1$ gives $k = \frac{1}{2}$.

Check: $P(X = x) = \frac{1}{2}\left(\frac{2}{3}\right)^x > 0$ for all $x \geq 1$, and the sum is 1. ✓

WARNING **Do not forget the check**

When the equation for k is quadratic, one root is almost always invalid because it produces a negative probability or a probability greater than 1. Examiners award a mark for explicitly rejecting the invalid root. Also, never assume k itself is a probability: in $P(X = x) = kx$, the constant k is just a scaling factor.

2.3 Computing probabilities of events from the distribution

Once the distribution is known, the probability of any event concerning X is found by **adding the masses** at the values that satisfy the event, because the events $\{X = x_i\}$ are mutually exclusive:

$$P(a \le X \le b) = \sum_{a \le x_i \le b} p(x_i).$$

EXEMPLE **Compound events**

Using the distribution $P(X = x) = 0.1x$ for $x = 1, 2, 3, 4$ found above:

- $P(X \le 2) = P(X = 1) + P(X = 2) = 0.1 + 0.2 = 0.3$;
- $P(X > 2) = P(X = 3) + P(X = 4) = 0.3 + 0.4 = 0.7$ (note: $= 1 - P(X \le 2)$);
- $P(2 \le X \le 3) = 0.2 + 0.3 = 0.5$;
- $P(X \text{ is even}) = P(X = 2) + P(X = 4) = 0.2 + 0.4 = 0.6$.

WARNING **Strict versus weak inequalities**

For a discrete variable, $P(X > 2)$ and $P(X \ge 2)$ are different: the first excludes $x = 2$, the second includes it. Read the inequality sign with great care — this is one of the most frequent sources of lost marks. (For continuous variables, met later, the two coincide.)

2.4 The discrete uniform distribution

Our first standard family of distributions is the simplest imaginable: all values equally likely.

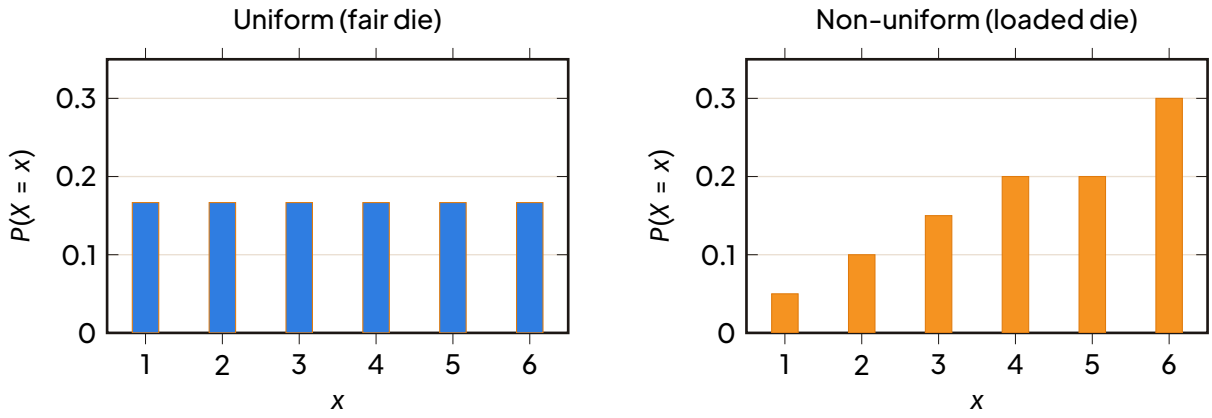
DEFINITION **Discrete uniform distribution**

A discrete random variable X has the **discrete uniform distribution** on $\{1, 2, \dots, n\}$ if

$$P(X = x) = \frac{1}{n} \quad \text{for } x = 1, 2, \dots, n.$$

This is a valid p.m.f. since each probability is positive and $n \times \frac{1}{n} = 1$.

The score on a fair die is uniform on $\{1, \dots, 6\}$; the number drawn in a fair tombola with tickets 1 to n is uniform on $\{1, \dots, n\}$. Uniformity is a property of the experiment, not of the set of values: the loaded die below has the same values but a very different distribution.



Two distributions on $\{1, 2, 3, 4, 5, 6\}$: the uniform distribution (every bar has height $1/6$) and a non-uniform distribution. In both cases the heights sum to 1.

2.5 Functions of a random variable: the distribution of $Y = g(X)$

Suppose X is the score on a die and you win $Y = 2X + 1$ francs, or you square the outcome, $Y = X^2$. Then Y is itself a discrete random variable, and we can derive its distribution from that of X. The principle is:

$$P(Y = y) = \sum_{x\,:\,g(x)=y} P(X = x),$$

that is, **add the probabilities of all values of X that are sent to the same value y**. When g is one-to-one on the values of X (like $g(x) = 2x + 1$), each y comes from a single x and the probabilities are simply carried over. When g is not one-to-one (like $g(x) = x^2$ when X takes negative values), several values of X may collapse onto one value of Y, and their probabilities must be added.

WORKED EXAMPLE 7 **Distribution of $Y = X^2$**

Let X have the distribution

x	-2	-1	0	1	2
$P(X = x)$	0.1	0.2	0.3	0.25	0.15

Find the probability distribution of $Y = X^2$.

SOLUTION

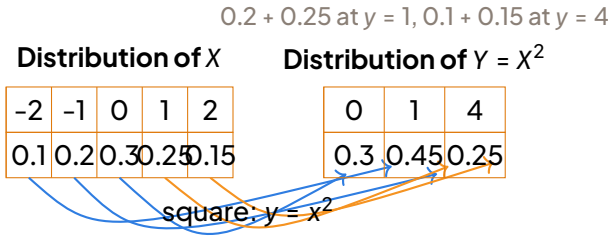
Solution. The possible values of Y are $0, 1, 4$. Grouping the values of X with the same square:

$$P(Y = 0) = P(X = 0) = 0.3,$$
$$P(Y = 1) = P(X = -1) + P(X = 1) = 0.2 + 0.25 = 0.45,$$
$$P(Y = 4) = P(X = -2) + P(X = 2) = 0.1 + 0.15 = 0.25.$$

Hence the distribution of Y is

y	0	1	4
$P(Y = y)$	0.3	0.45	0.25

Check: $0.3 + 0.45 + 0.25 = 1$. ✓



Transforming the distribution of X into that of $Y = X^2$: values of X with the same square merge, and their probabilities are added.

WARNING **Merging values under $Y = g(X)$**

When finding the distribution of $Y = X^2$, values of X with the same square (e.g. -2 and 2) must have their probabilities added. A very common error is to list $y = 4$ twice, or to copy only one of the two probabilities. Similarly, for $Y = |X|$ or $Y = (X - 1)^2$, always ask: which values of X give the same Y?

EXEMPLE **A one-to-one transformation: $Y = 2X + 1$**

If X is uniform on $\{1, 2, 3\}$ (each probability $\frac{1}{3}$), then $Y = 2X + 1$ takes the values $3, 5, 7$, each with probability $\frac{1}{3}$: the distribution of Y is uniform on $\{3, 5, 7\}$. No merging occurs because $g(x) = 2x + 1$ is strictly increasing.

EXEMPLE **A Cameroonian context: Loterie Nationale**

In a simplified version of the Loterie Nationale, a ticket bears a number X from 1 to 5 with probabilities:

x	1	2	3	4	5
$P(X = x)$	0.1	0.2	0.3	0.25	0.15

The prize is $Y = 1000X - 2000$ FCFA. Find the distribution of Y. Solution. $g(x) = 1000x - 2000$ is one-to-one. The values of Y are $-1000, 0, 1000, 2000, 3000$ with the same probabilities $0.1, 0.2, 0.3, 0.25, 0.15$ respectively.

EXERCISE 2 **Practice**

1. A random variable X has p.m.f. $P(X = x) = kx^2$ for $x = 1, 2, 3$. Find k, tabulate the distribution, and compute $P(X \geq 2)$ and $P(X \text{ is odd})$.
2. The distribution of X is given by $P(X = 0) = 0.5k$, $P(X = 1) = k$, $P(X = 2) = k^2$, $P(X = 3) = 0.1$. Find k.
3. X is uniform on $\{-2, -1, 0, 1, 2\}$. Find the distributions of $Y = X^2$ and of $Z = 3X - 1$.
4. A discrete random variable X takes values $0, 1, 2, 3$ with $P(X = x) = k(4 - x)$. Find k and the distribution of $Y = (X - 1)^2$.

SOLUTION **Exercise 1**

1. $\sum kx^2 = k(1 + 4 + 9) = 14k = 1$, so $k = \frac{1}{14}$. The probabilities are $\frac{1}{14}, \frac{4}{14}, \frac{9}{14}$ for $x = 1, 2, 3$. Then $P(X \geq 2) = \frac{4}{14} + \frac{9}{14} = \frac{13}{14}$ and $P(X \text{ odd}) = P(X = 1) + P(X = 3) = \frac{1}{14} + \frac{9}{14} = \frac{10}{14} = \frac{5}{7}$.

2. $0.5k + k + k^2 + 0.1 = 1 \Rightarrow k^2 + 1.5k - 0.9 = 0 \Rightarrow 10k^2 + 15k - 9 = 0$, giving $k = \dfrac{-15 \pm \sqrt{225 + 360}}{20} = \dfrac{-15 \pm \sqrt{585}}{20}$. Only the positive root is valid: $k = \dfrac{-15 + \sqrt{585}}{20} \approx 0.459$. (Check: probabilities $\approx 0.230, 0.459, 0.211, 0.1$, sum $= 1$.)

3. Each value of X has probability $\frac{1}{5} = 0.2$. For $Y = X^2$: $P(Y = 0) = 0.2$, $P(Y = 1) = 0.2 + 0.2 = 0.4$, $P(Y = 4) = 0.2 + 0.2 = 0.4$. For $Z = 3X - 1$: the values are $-7, -4, -1, 2, 5$, each with probability 0.2 (one-to-one map, no merging).

4. $\sum_{x=0}^{3} k(4 - x) = k(4 + 3 + 2 + 1) = 10k = 1 \Rightarrow k = 0.1$. Distribution of X:

x	0	1	2	3
$P(X = x)$	0.4	0.3	0.2	0.1

For $Y = (X - 1)^2$: $X = 0 \to Y = 1$, $X = 1 \to Y = 0$, $X = 2 \to Y = 1$, $X = 3 \to Y = 4$. $P(Y = 0) = P(X = 1) =$

0.3; $P(Y = 1) = P(X = 0) + P(X = 2) = 0.4 + 0.2 = 0.6$; $P(Y = 4) = P(X = 3) = 0.1$. Distribution of Y:

y	0	1	4
$P(Y = y)$	0.3	0.6	0.1

WARNING **Exam tip — GCE**

Always present the final distribution as a complete table showing every value of the variable and its probability. Marks are awarded for the full set of values and the full set of probabilities; a formula alone, or a table missing a value, loses marks. Finish with the check $\sum p_i = 1$ — it takes one line and often earns a method mark.

KEY POINTS **Summary of the section**

- The **probability mass function** of a discrete random variable is $p(x) = P(X = x)$.
- It must satisfy $p(x_i) \geq 0$ for all i and $\sum_i p(x_i) = 1$; the second condition is the tool for finding unknown constants.
- Probabilities of events are sums of masses: $P(a \leq X \leq b) = \sum_{a \leq x_i \leq b} p(x_i)$; watch strict versus weak inequalities.
- The **discrete uniform distribution** on $\{1, \ldots, n\}$ has $P(X = x) = \frac{1}{n}$.
- For $Y = g(X)$, $P(Y = y) = \sum_{g(x)=y} P(X = x)$: add probabilities of values of X that merge.

With the distribution of a discrete random variable now firmly in hand, we can ask the next natural question: how do we summarise a whole distribution by one or two numbers? That is the role of the expectation, variance and standard deviation, the subject of the next section.

3 EXPECTATION, VARIANCE AND STANDARD DEVIATION

In the previous section we learned how to describe a discrete random variable by its probability mass function (p.m.f.) or its probability distribution table. That table gives a complete picture of the possible values and their likelihoods. However, in practice we often need a single number that summarises the centre of the distribution (the "average" outcome we would expect in the long run) and another number that measures how much the values spread around that centre. These are the **expectation** (or mean) and the **variance** (with its square root, the standard deviation). They are the cornerstones of statistical decision-making — from evaluating a game of chance to comparing investment portfolios.

3.1 The expectation (mean) of a discrete random variable

Imagine a market trader in Douala who sells roasted plantains (beignets de plantain). Over many days she records her daily profit (in CFA francs). Some days she makes 5000, some days 10000, some days 2000, and occasionally she makes a loss. If we build the probability distribution of her daily profit X from a long history, the **expectation** $E(X)$ is exactly the long-run average profit per day. It is the value she can "expect" on a typical day, not because it will occur every day, but because the fluctuations balance out over time.

DEFINITION **Expectation of a discrete random variable**

Let X be a discrete random variable taking values $x_1, x_2, \dots, x_n$ with probabilities $p_1, p_2, \dots, p_n$ (where $p_i = P(X = x_i)$ and $\sum_{i=1}^{n} p_i = 1$). The expectation (or mean) of X, denoted $E(X)$ or μ, is defined by

$$E(X) = \sum_{i=1}^{n} x_i\, p_i = \sum_{i=1}^{n} x_i\, P(X = x_i).$$

It is the weighted average of the possible values, the weights being the probabilities.

EXEMPLE **Expectation of a fair die score**

Let X be the score when a fair six-sided die is rolled. The distribution is

x	1	2	3	4	5	6
$P(X = x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

Then

$$E(X) = \sum_{x=1}^{6} x \cdot \frac{1}{6} = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5.$$

Notice that 3.5 is not a possible outcome of a single roll; it is the theoretical centre of the distribution.

EXEMPLE **Expectation of a discrete uniform distribution**

If X takes values $1, 2, \dots, n$ each with probability $\frac{1}{n}$, then

$$E(X) = \frac{1}{n} \sum_{x=1}^{n} x = \frac{n(n+1)}{2n} = \frac{n+1}{2}.$$

For a fair die ($n = 6$), this gives $\frac{7}{2} = 3.5$ as before.

Expectation of a function of X. Often we need the average of a transformed variable, e.g. X^2 or $aX + b$. The same weighting principle applies:

PROPERTY **Expectation of a function**

For any real-valued function g,

$$E\big(g(X)\big) = \sum_i g(x_i)\,P(X = x_i).$$

In particular,

$$E(X^2) = \sum_i x_i^2\,P(X = x_i).$$

EXEMPLE **Computing $E(X^2)$ for the die**

Using the same die,

$$E(X^2) = \frac{1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2}{6} = \frac{91}{6} \approx 15.1667.$$

Linearity of expectation. One of the most powerful properties of expectation is that it is a **linear operator**. This means we can pull constants out and split sums — a fact that simplifies countless calculations.

PROPERTY **Linearity of expectation**

For any discrete random variable X and any constants $a, b \in \mathbb{R}$,

$$E(aX + b) = a\,E(X) + b.$$

More generally, for any two random variables X and Y (discrete or not, independent or not),

$$E(X + Y) = E(X) + E(Y).$$

PROPERTY **Proof of linearity**

Let X take values x_i with probabilities p_i. Then $aX + b$ takes values $ax_i + b$ with the same probabilities p_i. Hence

$$E(aX + b) = \sum_i (ax_i + b)\,p_i = a\sum_i x_i p_i + b\sum_i p_i$$
$$= a\,E(X) + b \cdot 1 = a\,E(X) + b.$$

The additivity $E(X + Y) = E(X) + E(Y)$ follows similarly by summing over the joint distribution (no independence required).

EXEMPLE **Linear transformation of the die score**

Suppose a game pays $Y = 2X + 100$ CFA francs, where X is the die score. Then

$$E(Y) = 2E(X) + 100 = 2 \times 3.5 + 100 = 107 \text{ CFA francs.}$$

We did not need to build the full distribution of Y; linearity did the work instantly.

EXEMPLE **Expectation of a non-linear function**

Let X be the score of a fair die. Compute $E(1/X)$.

$$E(1/X) = \sum_{x=1}^{6} \frac{1}{x} \cdot \frac{1}{6} = \frac{1}{6}\left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6}\right) = \frac{49}{120} \approx 0.4083.$$

Note that $E(1/X) \neq 1/E(X) = 2/7 \approx 0.2857$. Expectation does not commute with non-linear functions.

3.2 Variance and standard deviation

Expectation tells us where the distribution is centred, but not how spread out it is. Two distributions can have the same mean but very different variability. Consider two games:

- Game A: you win 1000 CFA with probability 0.5, lose 1000 CFA with probability 0.5. Mean = 0.
- Game B: you win 1,000,000 CFA with probability 0.5, lose 1,000,000 CFA with probability 0.5. Mean = 0.

Both have $E(X) = 0$, but Game B is enormously riskier. The **variance** quantifies this spread.

DEFINITION **Variance and standard deviation**

The variance of a discrete random variable X, denoted $\mathrm{Var}(X)$ or σ^2, is the expected squared deviation from the mean:

$$\mathrm{Var}(X) = E\big[(X - \mu)^2\big] = \sum_{i} (x_i - \mu)^2 \, p_i,$$

where $\mu = E(X)$. An equivalent and often more convenient formula is

$$\mathrm{Var}(X) = E(X^2) - \big[E(X)\big]^2.$$

The standard deviation is $\sigma = \sqrt{\mathrm{Var}(X)}$. It has the same units as X and measures the typical distance of outcomes from the mean.

PROPERTY **Proof of equivalence of variance formulas**

$$\begin{aligned} E\big[(X-\mu)^2\big] &= E(X^2 - 2\mu X + \mu^2) \\ &= E(X^2) - 2\mu E(X) + \mu^2 \\ &= E(X^2) - 2\mu^2 + \mu^2 = E(X^2) - \mu^2. \end{aligned}$$

EXEMPLE **Variance of a fair die score**

We already have $E(X) = 3.5$ and $E(X^2) = 91/6$. Hence

$$\operatorname{Var}(X) = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{91}{6} - \frac{49}{4} = \frac{182 - 147}{12} = \frac{35}{12} \approx 2.9167,$$

and $\sigma = \sqrt{35/12} \approx 1.708$.

EXEMPLE **Variance of a discrete uniform distribution**

For X uniform on $\{1, 2, \dots, n\}$, $E(X) = \frac{n+1}{2}$ and $E(X^2) = \frac{(n+1)(2n+1)}{6}$. Then

$$\operatorname{Var}(X) = \frac{(n+1)(2n+1)}{6} - \left(\frac{n+1}{2}\right)^2 = \frac{n^2 - 1}{12}.$$

For a fair die ($n = 6$), $\operatorname{Var}(X) = \frac{35}{12}$ as above.

Effect of linear transformations on variance. Adding a constant shifts the whole distribution but does not change its spread; multiplying by a constant stretches or compresses the spread.

PROPERTY **Variance of a linear transformation**

For any constants $a, b \in \mathbb{R}$,

$$\operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X).$$

Consequently, $\sigma_{aX+b} = |a|\,\sigma_X$.

PROPERTY **Proof**

Let $\mu = E(X)$. Then $E(aX + b) = a\mu + b$. Now

$$\begin{aligned}\operatorname{Var}(aX + b) &= E\big[(aX + b - (a\mu + b))^2\big] \\ &= E\big[a^2(X - \mu)^2\big] = a^2 E\big[(X - \mu)^2\big] = a^2 \operatorname{Var}(X).\end{aligned}$$

EXEMPLE **Variance of the game $Y = 2X + 100$**

From the die example, $\operatorname{Var}(X) = 35/12$. Then

$$\operatorname{Var}(Y) = 2^2 \times \frac{35}{12} = \frac{140}{12} = \frac{35}{3} \approx 11.667,$$

and $\sigma_Y = 2\sigma_X \approx 3.416$.

EXEMPLE **Variance of a shifted variable**

Let X be the die score and $Z = X - 3.5$ (centred variable). Then $E(Z) = 0$ and $\operatorname{Var}(Z) = \operatorname{Var}(X) = 35/12$. Shifting by the mean does not change variance.

3.3 Efficient computation: the working table

When computing $E(X)$ and $\operatorname{Var}(X)$ by hand, a structured table avoids errors and saves time.

METHOD **Working table for mean and variance**

1. List all possible values x_i in the first column.
2. Write the corresponding probabilities $p_i = P(X = x_i)$ in the second column.
3. Compute $x_i p_i$ (third column) and sum to get $E(X) = \sum x_i p_i$.
4. Compute $x_i^2 p_i$ (fourth column) and sum to get $E(X^2) = \sum x_i^2 p_i$.
5. Finally, $\operatorname{Var}(X) = E(X^2) - [E(X)]^2$ and $\sigma = \sqrt{\operatorname{Var}(X)}$.

EXEMPLE **Number of heads in three fair coin tosses**

Let X be the number of heads. The distribution is binomial with $n = 3, p = 0.5$:

x	0	1	2	3
p_i	1/8	3/8	3/8	1/8

Working table:

x_i	p_i	$x_i p_i$	$x_i^2 p_i$
0	1/8	0	0
1	3/8	3/8	3/8
2	3/8	6/8	12/8
3	1/8	3/8	9/8
Sum	1	12/8 = 1.5	24/8 = 3

Thus $E(X) = 1.5$, $E(X^2) = 3$, $\mathrm{Var}(X) = 3 - (1.5)^2 = 0.75$, $\sigma = \sqrt{0.75} \approx 0.866$.

EXEMPLE **A distribution with negative values**

A random variable X has distribution:

x	-2	-1	0	1	2
$P(X = x)$	0.1	0.2	0.4	0.2	0.1

Working table:

x_i	p_i	$x_i p_i$	$x_i^2 p_i$
-2	0.1	-0.2	0.4
-1	0.2	-0.2	0.2
0	0.4	0	0
1	0.2	0.2	0.2
2	0.1	0.2	0.4
Sum	1	0	1.2

$E(X) = 0$, $E(X^2) = 1.2$, $\mathrm{Var}(X) = 1.2$, $\sigma = \sqrt{1.2} \approx 1.095$.

3.4 Applications: fair games and decision making

A **fair game** is one in which the expected net gain is zero. If you pay an entry fee c to play a game that pays X, your net gain is $Y = X - c$. The game is fair iff $E(Y) = 0$, i.e. $c = E(X)$.

WORKED EXAMPLE 8 **Evaluating a game of chance**

A game consists of rolling a fair die. You receive CFA francs equal to the square of the score (i.e. X^2). The entry fee is 16 CFA. Is the game fair? If not, who has the advantage?

SOLUTION

Solution. We already computed $E(X^2) = 91/6 \approx 15.1667$ CFA. Net gain $Y = X^2 - 16$. Then $E(Y) = E(X^2) - 16 = 91/6 - 16 = (91 - 96)/6 = -5/6 \approx -0.833$ CFA. Since $E(Y) < 0$, the game is not fair; the player loses on average 0.83 CFA per game. The organiser has the advantage. To make it fair, the entry fee should be $E(X^2) = 91/6 \approx 15.17$ CFA.

WORKED EXAMPLE 9 **Finding the fair entry fee**

A bag contains 3 red balls and 2 blue balls. You draw one ball at random. If it is red you win 200 CFA; if blue you lose 100 CFA. What entry fee would make this a fair game?

SOLUTION

Solution. Let X be the net gain before entry fee. $P(\text{red}) = 3/5$, $P(\text{blue}) = 2/5$. $E(X) = 200 \times \frac{3}{5} + (-100) \times \frac{2}{5} = 120 - 40 = 80$ CFA. For a fair game, entry fee $c = E(X) = 80$ CFA.

Comparing two investments using mean and standard deviation. Suppose you must choose between two investment plans whose returns (in millions of CFA) are discrete random variables R_1 and R_2 with the following distributions:

Return	Plan A (R_1)	Plan B (R_2)
2	0.2	0.1
5	0.6	0.2
8	0.2	0.7

WORKED EXAMPLE 10 **Comparing Plan A and Plan B**

Compute $E(R)$ and σ_R for each plan and comment.

SOLUTION

Plan A:

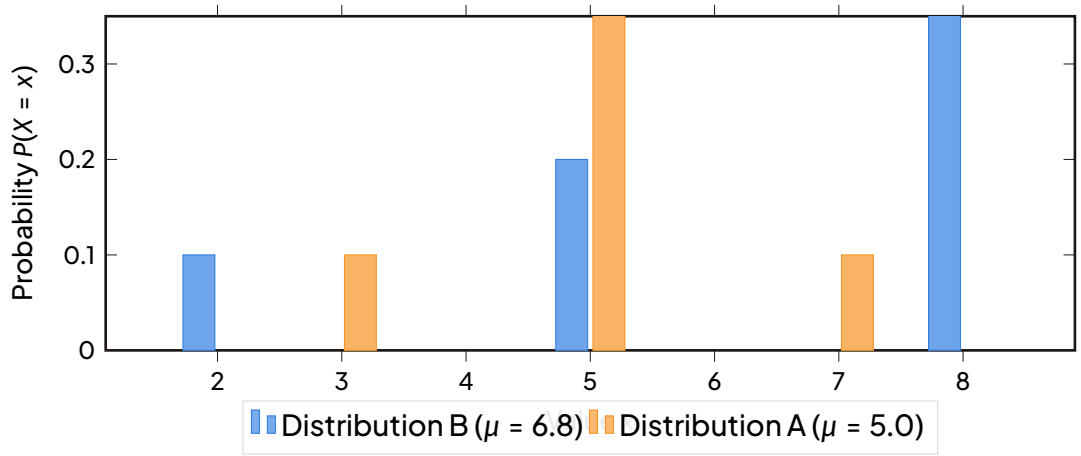
$$E(R_1) = 2(0.2) + 5(0.6) + 8(0.2) = 0.4 + 3.0 + 1.6 = 5.0$$
$$E(R_1^2) = 4(0.2) + 25(0.6) + 64(0.2) = 0.8 + 15.0 + 12.8 = 28.6$$
$$\text{Var}(R_1) = 28.6 - 5.0^2 = 3.6, \quad \sigma_1 = \sqrt{3.6} \approx 1.897.$$

Plan B:

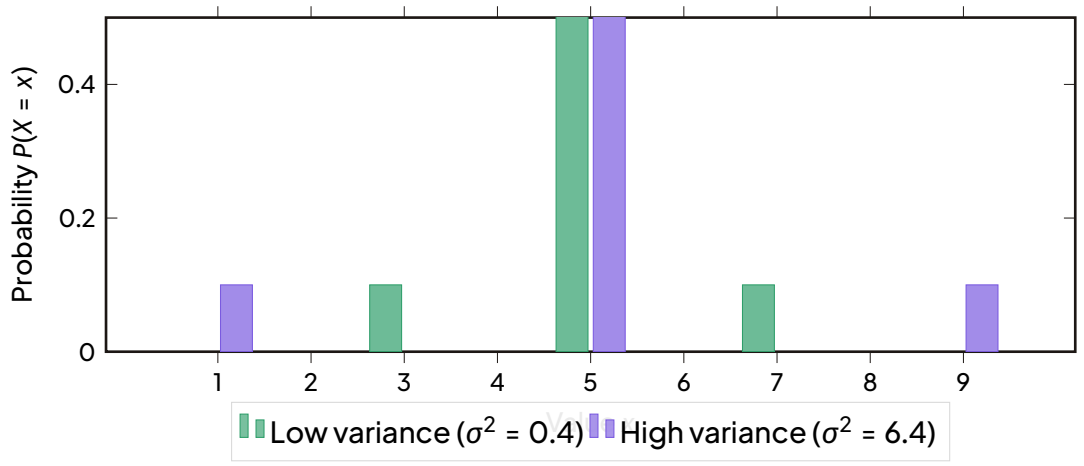
$$E(R_2) = 2(0.1) + 5(0.2) + 8(0.7) = 0.2 + 1.0 + 5.6 = 6.8$$
$$E(R_2^2) = 4(0.1) + 25(0.2) + 64(0.7) = 0.4 + 5.0 + 44.8 = 50.2$$
$$\mathrm{Var}(R_2) = 50.2 - 6.8^2 = 50.2 - 46.24 = 3.96, \quad \sigma_2 = \sqrt{3.96} \approx 1.990.$$

Interpretation: Plan B has a higher expected return (6.8 vs 5.0 million CFA) but also slightly higher risk (standard deviation 1.99 vs 1.90 million CFA). A risk-averse investor might prefer Plan A; a risk-tolerant one would choose Plan B. The decision depends on the investor's utility function, but E and σ provide the essential summary.

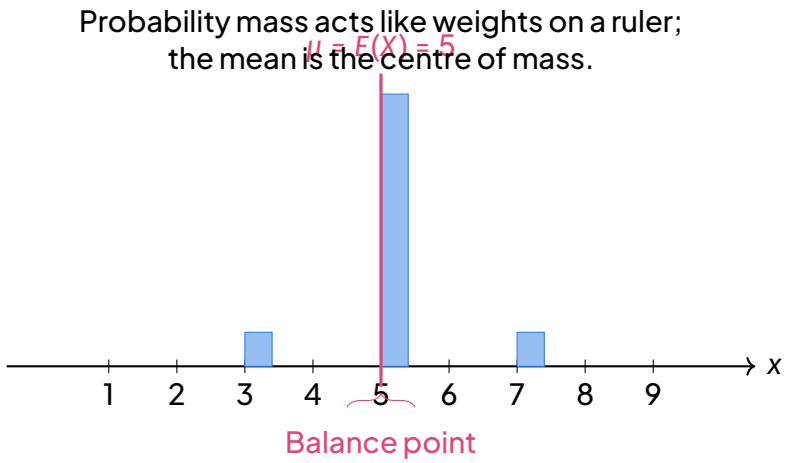
3.5 Illustrations



Two distributions with different means and variances. Distribution B has a higher mean but also a slightly larger spread.



Two distributions with the same mean ($\mu = 5$) but different variances. The purple distribution is much more spread out.



The expectation $E(X)$ is the balance point (centre of mass) of the probability bar chart. If the bars were physical weights on a light ruler, the ruler would balance exactly at $x = \mu$.

3.6 Common mistakes and warnings

WARNING **Variance formula traps**

1. Wrong: $\text{Var}(X) = E(X^2) - E(X)^2$ without brackets. The correct formula is $\text{Var}(X) = E(X^2) - [E(X)]^2$. The square is after taking the expectation of X, not before.

2. Wrong: $E(X^2) = [E(X)]^2$. This is false in general; it holds only when X is constant (variance zero). For the die, $E(X^2) = 91/6 \approx 15.17$ while $[E(X)]^2 = 12.25$.

3. Wrong: $\text{Var}(aX + b) = a\,\text{Var}(X) + b$. The correct rule is $\text{Var}(aX + b) = a^2\,\text{Var}(X)$. Adding a constant b does not change the variance; multiplying by a scales the variance by a^2.
4. Units: Variance is in squared units (e.g. CFA^2). Standard deviation returns to the original units (CFA). Always report σ when interpreting spread in context.
5. Wrong: $E(1/X) = 1/E(X)$. Expectation does not commute with non-linear functions (see example above).

WARNING **Examination tips**

- Always show your working table in GCE exams; it earns method marks even if arithmetic slips occur.
- When asked for variance, give the exact fraction (e.g. $35/12$) unless a decimal approximation is explicitly requested.
- For standard deviation, write $\sigma = \sqrt{\text{Var}(X)}$ and simplify the surd if possible (e.g. $\sqrt{0.75} = \sqrt{3}/2$).
- Check that probabilities sum to 1 before computing expectations.
- Remember: $E(X + Y) = E(X) + E(Y)$ always, but $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ only if X and Y are independent (covered in Lesson 138).

EXERCISE 3 **Practice: mean and variance of a discrete RV**

A random variable X has the following probability distribution:

x	-2	-1	0	1	2
$P(X = x)$	0.1	0.2	0.4	0.2	0.1

1. Verify that this is a valid probability distribution.
2. Compute $E(X)$ and $E(X^2)$.
3. Find $\text{Var}(X)$ and σ_X.
4. Let $Y = 3X - 5$. Write down the distribution of Y and compute $E(Y)$ and $\text{Var}(Y)$ using the linear transformation rules (check by direct calculation if you wish).

SOLUTION **Practice: mean and variance of a discrete RV**

1. Sum of probabilities = 0.1 + 0.2 + 0.4 + 0.2 + 0.1 = 1. Valid.
2. Working table:

x	p	xp	x^2p
-2	0.1	-0.2	0.4
-1	0.2	-0.2	0.2
0	0.4	0	0
1	0.2	0.2	0.2
2	0.1	0.2	0.4
Sum	1	0	1.2

   $E(X) = 0$, $E(X^2) = 1.2$.

3. $\mathrm{Var}(X) = 1.2 - 0^2 = 1.2$, $\sigma_X = \sqrt{1.2} = \sqrt{6/5} = \sqrt{30}/5 \approx 1.095$.
4. $Y = 3X - 5$. By linearity: $E(Y) = 3E(X) - 5 = -5$. $\mathrm{Var}(Y) = 3^2\,\mathrm{Var}(X) = 9 \times 1.2 = 10.8$. Direct check: Y takes values -11, -8, -5, -2, 1 with same probabilities. $E(Y) = -11(0.1) - 8(0.2) - 5(0.4) - 2(0.2) + 1(0.1) = -1.1 - 1.6 - 2.0 - 0.4 + 0.1 = -5.0$. OK. $E(Y^2) = 121(0.1) + 64(0.2) + 25(0.4) + 4(0.2) + 1(0.1) = 12.1 + 12.8 + 10.0 + 0.8 + 0.1 = 35.8$. $\mathrm{Var}(Y) = 35.8 - (-5)^2 = 35.8 - 25 = 10.8$. OK.

EXERCISE 4 **GCE-style question**

The discrete random variable X has probability distribution given by

$$P(X = x) = \frac{x}{10}, \quad x = 1, 2, 3, 4.$$

1. Show that this is a valid probability distribution.
2. Find $E(X)$ and $\mathrm{Var}(X)$.
3. Given that $Y = 2X - 3$, find $E(Y)$ and $\mathrm{Var}(Y)$.

SOLUTION **GCE-style question**

1. $\sum_{x=1}^{4} P(X = x) = \frac{1+2+3+4}{10} = \frac{10}{10} = 1$. Valid.

2. Working table:

x	$p = x/10$	xp	x^2p
1	0.1	0.1	0.1
2	0.2	0.4	0.8
3	0.3	0.9	2.7
4	0.4	1.6	6.4
Sum	1	3.0	10.0

$E(X) = 3$, $E(X^2) = 10$, $\text{Var}(X) = 10 - 9 = 1$.

3. $E(Y) = 2E(X) - 3 = 2(3) - 3 = 3$. $\text{Var}(Y) = 2^2\,\text{Var}(X) = 4 \times 1 = 4$.

KEY POINTS **Key points of this section**

- $E(X) = \sum x_i P(X = x_i)$ is the theoretical mean (long-run average).
- $E(g(X)) = \sum g(x_i)P(X = x_i)$; in particular $E(X^2) = \sum x_i^2 p_i$.
- Linearity: $E(aX + b) = aE(X) + b$ and $E(X + Y) = E(X) + E(Y)$ (always, no independence needed).
- $\text{Var}(X) = E(X^2) - [E(X)]^2 \geq 0$; $\sigma = \sqrt{\text{Var}(X)}$.
- $\text{Var}(aX + b) = a^2\,\text{Var}(X)$; adding a constant does not affect variance.
- Use a working table (x, p, xp, x^2p) for accurate hand calculations.
- A fair game has $E(\text{net gain}) = 0$; fair entry fee = $E(\text{payout})$.
- When comparing options, consider both $E(X)$ (return) and σ (risk).
- $E(X^2) \neq [E(X)]^2$ unless $\text{Var}(X) = 0$.
- For independent X, Y (later lesson): $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.

4 THE CUMULATIVE DISTRIBUTION FUNCTION, MODE AND MEDIAN

In the previous section we summarised a discrete distribution by a single number, its **expectation** $E(X)$, and measured its spread with the **variance** and **standard deviation**. These are excellent summaries, but they do not tell the whole story. Two distributions can share the same mean yet look completely different. To describe a distribution fully, we need a function that accumulates probabilities as we move along the number line: the **cumulative distribution func-**

tion. From it we can also read off two other important measures of location, the **mode** and the **median**, and compare them with the mean.

4.1 The cumulative distribution function

Motivation.

Suppose a trader at Mokolo market records the number X of bags of rice she sells per day, with distribution

x_i	0	1	2	3
$P(X = x_i)$	0.1	0.3	0.4	0.2

Questions such as "what is the probability that she sells at most 2 bags?" arise constantly. The answer is $P(X \le 2) = 0.1 + 0.3 + 0.4 = 0.8$. Rather than recomputing such sums every time, we build a single function that stores all these running totals.

DEFINITION **Cumulative distribution function**

Let X be a discrete random variable. The **cumulative distribution function** (c.d.f.) of X is the function F defined for every real number x by

$$F(x) = P(X \le x).$$

Constructing F from the probability table.

For a distribution taking values $x_1 < x_2 < \cdots < x_n$, the c.d.f. is built by **cumulative addition** of the probabilities:

- for $x < x_1$: $F(x) = 0$;
- for $x_1 \le x < x_2$: $F(x) = P(X = x_1)$;
- for $x_2 \le x < x_3$: $F(x) = P(X = x_1) + P(X = x_2)$;
- …
- for $x \ge x_n$: $F(x) = 1$.

For our trader, this gives

$$F(x) = \begin{cases} 0 & \text{if } x < 0, \\ 0.1 & \text{if } 0 \le x < 1, \\ 0.4 & \text{if } 1 \le x < 2, \\ 0.8 & \text{if } 2 \le x < 3, \\ 1 & \text{if } x \ge 3. \end{cases}$$

WARNING **F is defined for ALL real x**

A frequent GCE trap: $F(x)$ is defined for every real number x, not only at the values x_i. Between two consecutive values, F stays constant. For instance, $F(1.7) = P(X \le 1.7) = P(X = 0) + P(X = 1) = 0.4$, and $F(-5) = 0$. Never leave gaps in the definition of F.

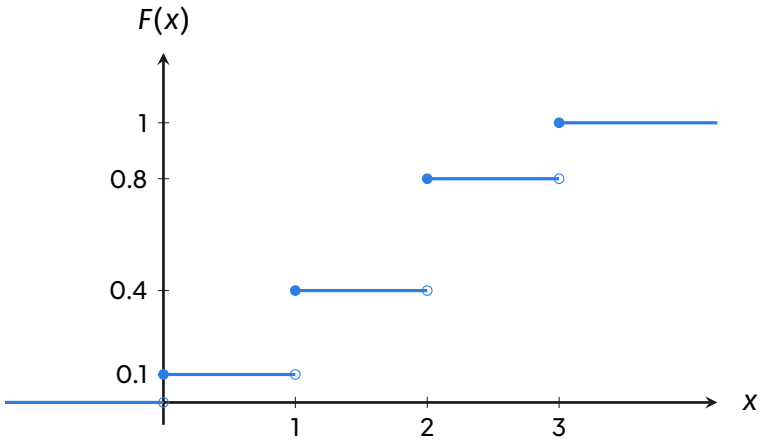
PROPERTY **Properties of the c.d.f.**

For any random variable X with c.d.f. F:

1. $0 \le F(x) \le 1$ for all real x;
2. F is **non-decreasing**: if $a \le b$ then $F(a) \le F(b)$;
3. $F(x) \to 0$ as $x \to -\infty$ and $F(x) \to 1$ as $x \to +\infty$;
4. F is **right-continuous**: at each value x_i the function jumps, and the value at the jump is the upper value, $F(x_i) = P(X \le x_i)$.

(The proof of these properties is not examinable, but you must know and use them.)

Justification of (2). If $a \le b$, the event $\{X \le a\}$ is contained in the event $\{X \le b\}$, so $P(X \le a) \le P(X \le b)$, that is $F(a) \le F(b)$. □



The c.d.f. $F(x)$ of the rice-sales distribution: a step function with jumps at $x = 0, 1, 2, 3$. Closed dots show the value taken at each jump (right-continuity).

Recovering probabilities from F.

The c.d.f. contains all the information of the distribution, so we can travel back from F to the individual probabilities.

PROPERTY **From the c.d.f. back to probabilities**

If X takes values $x_1 < x_2 < \cdots < x_n$, then

$$P(X = x_i) = F(x_i) - F(x_{i-1}) \qquad (i \geq 2), \quad P(X = x_1) = F(x_1),$$

and for any real numbers $a < b$,

$$P(a < X \leq b) = F(b) - F(a).$$

Justification. The event $\{X \leq b\}$ is the disjoint union of $\{X \leq a\}$ and $\{a < X \leq b\}$, so $F(b) = F(a) + P(a < X \leq b)$, which gives the second formula. Taking $a = x_{i-1}$ and $b = x_i$ gives the first, since the only value of X in $(x_{i-1}, x_i]$ is x_i. □

WARNING **Strict versus weak inequalities**

Because X is discrete, $P(a < X \leq b) = F(b) - F(a)$ but $P(a \leq X \leq b) = F(b) - F(a) + P(X = a)$ when a is one of the values. Always check whether the endpoints are included.

WORKED EXAMPLE 11 **Reading a c.d.f.**

A discrete random variable X has c.d.f.

$$F(x) = \begin{cases} 0 & x < 1, \\ 0.2 & 1 \leq x < 2, \\ 0.5 & 2 \leq x < 4, \\ 0.9 & 4 \leq x < 5, \\ 1 & x \geq 5. \end{cases}$$

Find (a) the probability distribution of X; (b) $P(2 < X \leq 5)$; (c) $P(X \geq 4)$.

SOLUTION

Solution. (a) The jumps of F occur at $x = 1, 2, 4, 5$, so these are the values of X, and each probability is the size of the jump:

$$P(X = 1) = 0.2 - 0 = 0.2, \qquad P(X = 2) = 0.5 - 0.2 = 0.3,$$
$$P(X = 4) = 0.9 - 0.5 = 0.4, \qquad P(X = 5) = 1 - 0.9 = 0.1.$$

Check: $0.2 + 0.3 + 0.4 + 0.1 = 1$. ✓
(b) $P(2 < X \leq 5) = F(5) - F(2) = 1 - 0.5 = 0.5$ (indeed $P(X = 4) + P(X = 5) = 0.5$).
(c) $P(X \geq 4) = 1 - P(X \leq 2) = 1 - F(2) = 1 - 0.5 = 0.5$. Note we subtract $F(2)$, not $F(4)$, because the event $\{X \geq 4\}$ is the complement of $\{X \leq 2\}$ (there is no value 3).

4.2 The mode

Intuition.

The **mode** answers the question: which outcome is the most likely? It is simply the value of X that carries the greatest probability — the tallest bar of the bar chart.

DEFINITION **Mode of a discrete distribution**

A **mode** of a discrete random variable X is a value x_i such that

$$P(X = x_i) \geq P(X = x_j) \quad \text{for every value } x_j.$$

A distribution with one mode is called **unimodal**; if two values share the greatest probability, the distribution is **bimodal**.

For the rice-sales distribution, $P(X = 2) = 0.4$ is the largest probability, so the mode is 2 bags. If instead the probabilities had been 0.1, 0.4, 0.4, 0.1, the distribution would be bimodal with modes 1 and 2.

4.3 The median

Intuition.

The **median** is the "half-way point" of the distribution: a value m such that at least half of the probability lies at or below m. For a discrete variable we must phrase this carefully using the c.d.f.

DEFINITION **Median of a discrete distribution**

The **median** m of a discrete random variable X is the smallest value of X such that

$$F(m) = P(X \leq m) \geq 0.5.$$

METHOD **Finding the median from a probability table**

1. Write the values $x_1 < x_2 < \ldots$ in increasing order.
2. Compute the cumulative probabilities $F(x_1), F(x_2), \ldots$ by adding the probabilities from the left.
3. Stop at the first value x_i where the cumulative total reaches or exceeds 0.5: that value is the median.

WARNING **Do not interpolate!**

The median of a discrete distribution is always one of the actual values of X. It is not necessary that $F(m)$ equals exactly 0.5. If $F(1) = 0.4$ and $F(2) = 0.8$, the median is 2 — do not invent an intermediate value such as 1.25 by interpolation. Interpolation is a mistake the GCE examiners penalise every year.

For the rice-sales distribution: $F(0) = 0.1$, $F(1) = 0.4$, $F(2) = 0.8$. The first cumulative total reaching 0.5 is $F(2) = 0.8$, so the median is $m = 2$.

4.4 Comparing mean, mode and median

These three numbers all describe the "centre" of a distribution, but from different points of view:

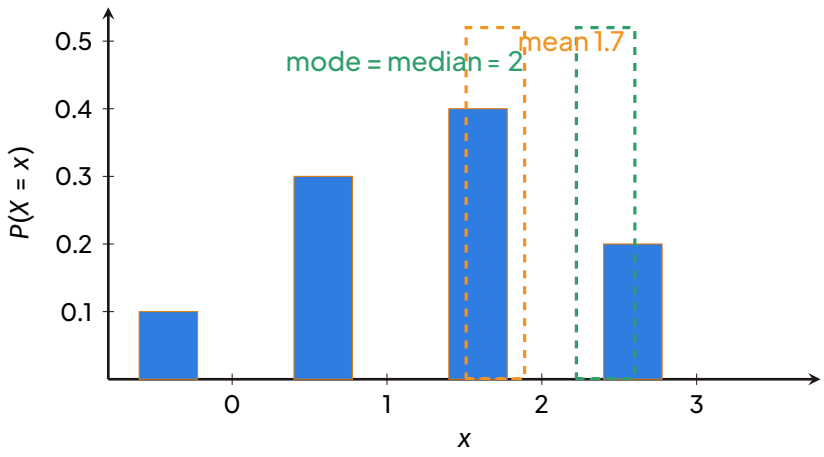
- the **mode** is the most probable value;
- the **median** splits the distribution so that at least half the probability lies on each side;
- the **mean** $E(X)$ is the balance point, taking every value into account weighted by its probability.

EXEMPLE **Mean, mode and median of one distribution**

For the rice-sales distribution:

$$E(X) = 0(0.1) + 1(0.3) + 2(0.4) + 3(0.2) = 1.7.$$

So we have: mode = 2, median = 2, mean = 1.7. The mean is pulled slightly below the mode and median because the value 0 drags the balance point to the left, while the median and mode, which ignore the sizes of the values and look only at probabilities, stay at 2.



Bar chart of the distribution with the mean (orange dashed line) and the common value of the mode and median (green dashed line).

DID YOU KNOW? **Why the median matters in Cameroon**

In many real-life situations in Cameroon — household sizes in Yaoundé, daily earnings of moto-taxi drivers in Douala, or cocoa yields per hectare in the South-West — the distribution is often skewed. The mean can be distorted by a few very large values, while the median gives a more representative "typical" value. That is why government reports on living standards often quote the median income rather than the mean.

WORKED EXAMPLE 12 **A complete study**

A random variable X has distribution

x_i	0	1	2	3	4
$P(X = x_i)$	0.05	0.25	0.4	0.2	0.1

(a) Construct the c.d.f. F. (b) Find the mode and the median. (c) Compute $E(X)$ and comment. (d) Find $P(1 < X \le 3)$.

SOLUTION

Solution. (a) Cumulative sums: $F(0) = 0.05$, $F(1) = 0.30$, $F(2) = 0.70$, $F(3) = 0.90$, $F(4) = 1$.

Hence

$$F(x) = \begin{cases} 0 & x < 0, \\ 0.05 & 0 \le x < 1, \\ 0.30 & 1 \le x < 2, \\ 0.70 & 2 \le x < 3, \\ 0.90 & 3 \le x < 4, \\ 1 & x \ge 4. \end{cases}$$

(b) The greatest probability is $P(X = 2) = 0.4$, so the mode is 2. The first cumulative total reaching 0.5 is $F(2) = 0.70$, so the median is 2.
(c) $E(X) = 0(0.05) + 1(0.25) + 2(0.4) + 3(0.2) + 4(0.1) = 0.25 + 0.8 + 0.6 + 0.4 = 2.05$. The mean (2.05), median (2) and mode (2) are very close, which reflects the fact that the distribution is nearly symmetric about $x = 2$.
(d) $P(1 < X \le 3) = F(3) - F(1) = 0.90 - 0.30 = 0.60$ (check: $P(X = 2) + P(X = 3) = 0.4 + 0.2 = 0.6$). ✓

KEY POINTS **Key points of this section**

- The c.d.f. is $F(x) = P(X \le x)$, defined for every real x; it is a non-decreasing step function, right-continuous, with $F(x) \to 0$ as $x \to -\infty$ and $F(x) \to 1$ as $x \to +\infty$.
- Probabilities are recovered from F: $P(X = x_i) = F(x_i) - F(x_{i-1})$ and $P(a < X \le b) = F(b) - F(a)$.
- The **mode** is the value with the greatest probability (a distribution may be bimodal).
- The **median** is the smallest value m with $F(m) \ge 0.5$; it is always an actual value of X — never interpolate.
- Mean, median and mode coincide for a symmetric distribution but differ for skewed ones.

EXERCISE 5 **Practice**

A discrete random variable X takes values 1, 2, 3, 4 with probabilities 0.3, 0.3, 0.2, 0.2 respectively.

1. Write down the c.d.f. $F(x)$ as a piecewise function and sketch it.
2. State the mode(s) of X.
3. Find the median of X.
4. Compute $E(X)$ and compare the three measures of location.

5. Evaluate $P(1 < X \le 3)$ and $P(X \ge 2)$ using F.

SOLUTION **Exercise 1**

1. Cumulative sums: $F(1) = 0.3$, $F(2) = 0.6$, $F(3) = 0.8$, $F(4) = 1$. So

$$F(x) = \begin{cases} 0 & x < 1, \\ 0.3 & 1 \le x < 2, \\ 0.6 & 2 \le x < 3, \\ 0.8 & 3 \le x < 4, \\ 1 & x \ge 4. \end{cases}$$

The graph is a step function with jumps of heights 0.3, 0.3, 0.2, 0.2 at $x = 1, 2, 3, 4$.
2. The greatest probability is 0.3, attained at both $x = 1$ and $x = 2$: the distribution is bimodal with modes 1 and 2.
3. $F(1) = 0.3 < 0.5$ and $F(2) = 0.6 \ge 0.5$, so the median is $m = 2$.
4. $E(X) = 1(0.3)+2(0.3)+3(0.2)+4(0.2) = 0.3+0.6+0.6+0.8 = 2.3$. The mean (2.3) exceeds the median (2): the larger probabilities on the small values are balanced by the heavier values 3 and 4, which pull the balance point to the right.
5. $P(1 < X \le 3) = F(3) - F(1) = 0.8 - 0.3 = 0.5$; $P(X \ge 2) = 1 - F(1) = 1 - 0.3 = 0.7$.

5 THE DISTRIBUTION OF X₁ + X₂

In the previous section we studied how to summarise a single discrete random variable using its cumulative distribution function, its mode and its median. Many real-world situations, however, involve the sum of two or more random variables: the total score when two dice are thrown, the total number of customers arriving at a shop over two consecutive days, the combined weight of two randomly chosen bags of rice, etc. This section develops the tools to obtain the probability distribution of a sum $S = X_1 + X_2$, and to compute its expectation and variance. The results here are the foundation for the binomial distribution (sum of n independent Bernoulli trials) and for the Central Limit Theorem later in the course.

5.1 Why sums of random variables?

EXPERIMENT **Total score of two dice**

Suppose we roll two fair six-sided dice, one red and one blue. Let X_1 be the score on the red die and X_2 the score on the blue die. Both are discrete uniform on $\{1, 2, 3, 4, 5, 6\}$. We are

often interested in the total score $S = X_1 + X_2$. The possible values of S range from 2 to 12, but they are not equally likely. For instance, $S = 7$ can occur in six ways $(1,6), (2,5), \ldots, (6,1)$ while $S = 2$ occurs only as $(1,1)$. To find the distribution of S we must combine the distributions of X_1 and X_2 in a systematic way.

EXPERIMENT **Total customers over two days**

A small café in Yaoundé records the number of customers each day. Let X_1 be the number on Monday and X_2 the number on Tuesday. If the daily numbers are independent and each follows a known distribution (say Poisson with mean 5), the owner may want the distribution of the two-day total $S = X_1 + X_2$ to plan staffing and supplies.

5.2 Independence of two discrete random variables

Before we can combine distributions we need a precise notion of independence for random variables.

DEFINITION **Independence of two discrete random variables**

Two discrete random variables X_1 and X_2 are **independent** if and only if for every possible value x of X_1 and every possible value y of X_2,

$$P(X_1 = x \text{ and } X_2 = y) = P(X_1 = x) \cdot P(X_2 = y).$$

Equivalently, the joint probability mass function factorises: $p_{X_1,X_2}(x,y) = p_{X_1}(x)\,p_{X_2}(y)$ for all x, y.

WARNING **Independence vs. uncorrelated**

Independence is a stronger condition than zero correlation. If X_1 and X_2 are independent then they are uncorrelated, but the converse is false in general. In the GCE syllabus we only use the definition above; always check whether the problem states or implies independence before using the product rule for joint probabilities.

5.3 Constructing the distribution of $S = X_1 + X_2$ by enumeration

When X_1 and X_2 are independent, the most transparent way to obtain the distribution of $S = X_1 + X_2$ is to build a **two-way table** (also called a joint probability table).

METHOD **Two-way table for the sum of two independent discrete random variables**

1. List the possible values of X_1 down the rows and those of X_2 across the columns.
2. In each cell write the sum $x_1 + x_2$ (the value of S) and the joint probability $P(X_1 = x_1)P(X_2 = x_2)$.
3. Collect all cells that give the same sum s; add their probabilities to obtain $P(S = s)$.
4. Present the result as a probability distribution table for S.

Let us apply this to the two-dice example.

WORKED EXAMPLE 13 **Sum of two fair dice**

Let X_1 and X_2 be the scores on two independent fair dice. Find the probability distribution of $S = X_1 + X_2$.

SOLUTION

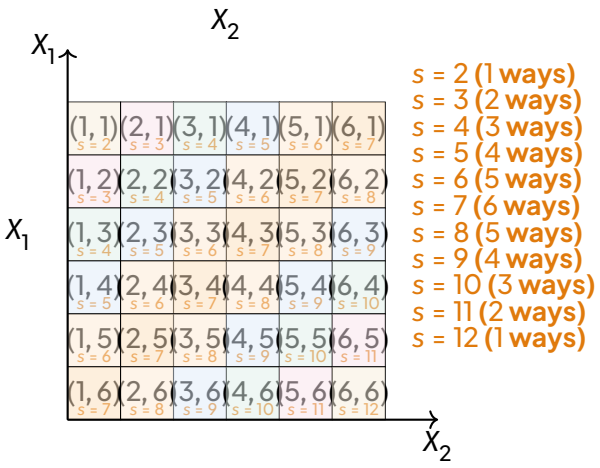
Step 1: Two-way table. Each die has outcomes $1, \dots, 6$ with probability $\frac{1}{6}$. The joint probability of any pair (x_1, x_2) is $\frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$.

$X_1 \backslash X_2$	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

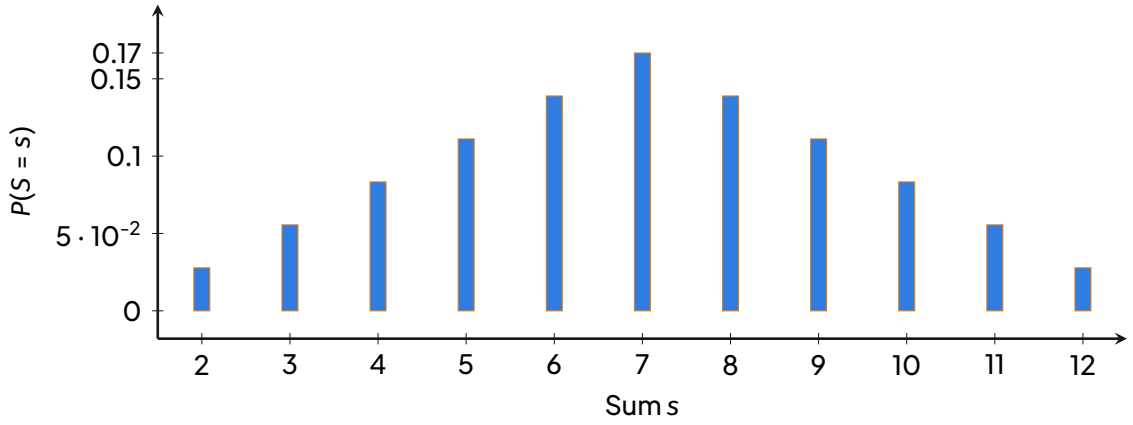
Each cell has probability $\frac{1}{36}$. The diagonals (constant $x_1 + x_2$) give the values of S.
Step 2: Collect probabilities along diagonals.

s	2	3	4	5	6	7	8	9	10	11	12
$P(S = s)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Step 3: Simplify (optional). $P(S = s) = \frac{6 - |s-7|}{36}$ for $s = 2, \dots, 12$.
This is the classic triangular distribution.



The 36 equally likely outcomes of two fair dice. Cells with the same sum s lie on diagonals; the number of cells on each diagonal gives the frequency of that sum.



Probability mass function of the sum of two fair dice — a symmetric triangular distribution peaking at $s = 7$.

5.4 The convolution formula

The two-way table method is perfectly general. For independent X_1 and X_2 with probability mass functions $p_1(x) = P(X_1 = x)$ and $p_2(y) = P(X_2 = y)$, the probability that $S = X_1 + X_2$ takes the value s is obtained by summing the joint probabilities over all pairs (x, y) with $x + y = s$. This is the **convolution** of the two distributions.

PROPERTY **Convolution formula for the sum of two independent discrete random variables**

If X_1 and X_2 are independent discrete random variables with probability mass functions p_1

and p_2, then for any integer s,

$$P(S = s) = P(X_1 + X_2 = s) = \sum_x p_1(x)\, p_2(s - x),$$

where the sum runs over all x such that $p_1(x) > 0$ and $p_2(s - x) > 0$.

EXEMPLE **Convolution of two independent uniform variables on** $\{1, 2, 3\}$

Let X_1 and X_2 be independent, each uniformly distributed on $\{1, 2, 3\}$ (so $p_1(x) = p_2(x) = \frac{1}{3}$ for $x = 1, 2, 3$). Find the distribution of $S = X_1 + X_2$.
The possible sums are $s = 2, 3, 4, 5, 6$. Using the convolution formula:

$$P(S = 2) = p_1(1)p_2(1) = \tfrac{1}{3} \cdot \tfrac{1}{3} = \tfrac{1}{9},$$
$$P(S = 3) = p_1(1)p_2(2) + p_1(2)p_2(1) = \tfrac{1}{9} + \tfrac{1}{9} = \tfrac{2}{9},$$
$$P(S = 4) = p_1(1)p_2(3) + p_1(2)p_2(2) + p_1(3)p_2(1) = \tfrac{1}{9} + \tfrac{1}{9} + \tfrac{1}{9} = \tfrac{3}{9},$$
$$P(S = 5) = p_1(2)p_2(3) + p_1(3)p_2(2) = \tfrac{1}{9} + \tfrac{1}{9} = \tfrac{2}{9},$$
$$P(S = 6) = p_1(3)p_2(3) = \tfrac{1}{9}.$$

The distribution is again triangular: $\frac{1}{9}, \frac{2}{9}, \frac{3}{9}, \frac{2}{9}, \frac{1}{9}$.

5.5 Expectation and variance of a sum

One of the most powerful properties of expectation is its **linearity** — it holds without any independence assumption.

PROPERTY **Expectation of a sum**

For any two discrete random variables X_1 and X_2 (independent or not),

$$E(X_1 + X_2) = E(X_1) + E(X_2).$$

More generally, $E\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} E(X_i)$.

Proof.

$$\begin{aligned} E(X_1 + X_2) &= \sum_x \sum_y (x + y)\, P(X_1 = x, X_2 = y) \\ &= \sum_x \sum_y x\, P(X_1 = x, X_2 = y) + \sum_x \sum_y y\, P(X_1 = x, X_2 = y) \\ &= \sum_x x\Big(\sum_y P(X_1 = x, X_2 = y)\Big) + \sum_y y\Big(\sum_x P(X_1 = x, X_2 = y)\Big) \\ &= \sum_x x\, P(X_1 = x) + \sum_y y\, P(X_2 = y) = E(X_1) + E(X_2). \end{aligned}$$

No independence was used; only the law of total probability for the marginals. □

For variance, independence is essential.

PROPERTY **Variance of a sum of independent random variables**

If X_1 and X_2 are **independent**, then

$$\operatorname{Var}(X_1 + X_2) = \operatorname{Var}(X_1) + \operatorname{Var}(X_2).$$

If X_1 and X_2 are not independent, this formula does not hold in general; one must use $\operatorname{Var}(X_1 + X_2) = \operatorname{Var}(X_1) + \operatorname{Var}(X_2) + 2\operatorname{Cov}(X_1, X_2)$ (covariance is beyond the GCE syllabus).

Sketch (examinable). $\operatorname{Var}(X_1 + X_2) = E[(X_1 + X_2)^2] - [E(X_1 + X_2)]^2$. Expand $(X_1 + X_2)^2 = X_1^2 + 2X_1X_2 + X_2^2$. By linearity of expectation, $E[(X_1 + X_2)^2] = E(X_1^2) + 2E(X_1X_2) + E(X_2^2)$. Independence gives $E(X_1X_2) = E(X_1)E(X_2)$. Then

$$\begin{aligned} \operatorname{Var}(X_1 + X_2) &= E(X_1^2) - E(X_1)^2 + E(X_2^2) - E(X_2)^2 \\ &= \operatorname{Var}(X_1) + \operatorname{Var}(X_2). \end{aligned}$$

□

WARNING **Variances add, standard deviations do NOT**

A very common mistake is to write $\sigma_{X_1+X_2} = \sigma_{X_1} + \sigma_{X_2}$. This is false. The correct rule for independent variables is

$$\sigma^2_{X_1+X_2} = \sigma^2_{X_1} + \sigma^2_{X_2} \quad \Longrightarrow \quad \sigma_{X_1+X_2} = \sqrt{\sigma^2_{X_1} + \sigma^2_{X_2}}.$$

Always work with variances when combining independent random variables; take the square root only at the very end if a standard deviation is required.

5.6 Worked examples: checking the rules

WORKED EXAMPLE 14 **Sum of two fair dice — direct vs. rules**

Let X_1, X_2 be independent fair dice scores. We already have the distribution of $S = X_1 + X_2$. Verify that $E(S) = E(X_1) + E(X_2)$ and $\mathrm{Var}(S) = \mathrm{Var}(X_1) + \mathrm{Var}(X_2)$.

SOLUTION

For a single die: $E(X_i) = \frac{1+2+3+4+5+6}{6} = 3.5$. $E(X_i^2) = \frac{1^2+2^2+3^2+4^2+5^2+6^2}{6} = \frac{91}{6} \approx 15.1667$. $\mathrm{Var}(X_i) = \frac{91}{6} - (3.5)^2 = \frac{91}{6} - \frac{49}{4} = \frac{182-147}{12} = \frac{35}{12} \approx 2.9167$.
By the rules: $E(S) = 3.5 + 3.5 = 7$. $\mathrm{Var}(S) = \frac{35}{12} + \frac{35}{12} = \frac{35}{6} \approx 5.8333$.
Directly from the distribution of S:

$$E(S) = \sum_{s=2}^{12} s\,P(S = s)$$
$$= \frac{1}{36}\big(2\cdot 1 + 3\cdot 2 + 4\cdot 3 + 5\cdot 4 + 6\cdot 5 + 7\cdot 6 + 8\cdot 5 + 9\cdot 4 + 10\cdot 3 + 11\cdot 2 + 12\cdot 1\big)$$
$$= \frac{1}{36}(2 + 6 + 12 + 20 + 30 + 42 + 40 + 36 + 30 + 22 + 12) = \frac{252}{36} = 7. \checkmark$$

$E(S^2) = \frac{1}{36}\big(4\cdot 1 + 9\cdot 2 + 16\cdot 3 + 25\cdot 4 + 36\cdot 5 + 49\cdot 6 + 64\cdot 5 + 81\cdot 4 + 100\cdot 3 + 121\cdot 2 + 144\cdot 1\big) = \frac{1974}{36} = \frac{329}{6} \approx 54.8333$. $\mathrm{Var}(S) = \frac{329}{6} - 49 = \frac{329-294}{6} = \frac{35}{6}$. ✓
Both methods agree perfectly.

WORKED EXAMPLE 15 **Two independent uniform variables on $\{0, 1, 2\}$**

Let X_1 and X_2 be independent, each with distribution $P(X = 0) = 0.2$, $P(X = 1) = 0.5$, $P(X = 2) = 0.3$. Find the distribution of $S = X_1 + X_2$, then compute $E(S)$ and $\mathrm{Var}(S)$ both directly and by the rules.

SOLUTION

Distribution of S (convolution): Possible sums $s = 0, 1, 2, 3, 4$.

$$P(S = 0) = 0.2 \times 0.2 = 0.04,$$
$$P(S = 1) = 0.2 \times 0.5 + 0.5 \times 0.2 = 0.20,$$
$$P(S = 2) = 0.2 \times 0.3 + 0.5 \times 0.5 + 0.3 \times 0.2 = 0.06 + 0.25 + 0.06 = 0.37,$$
$$P(S = 3) = 0.5 \times 0.3 + 0.3 \times 0.5 = 0.30,$$
$$P(S = 4) = 0.3 \times 0.3 = 0.09.$$

Check: $0.04 + 0.20 + 0.37 + 0.30 + 0.09 = 1.00$.
Expectation and variance of one variable: $E(X) = 0 \times 0.2 + 1 \times 0.5 + 2 \times 0.3 = 1.1$. $E(X^2) = 0^2 \times 0.2 + 1^2 \times 0.5 + 2^2 \times 0.3 = 0.5 + 1.2 = 1.7$. $\mathrm{Var}(X) = 1.7 - 1.1^2 = 1.7 - 1.21 = 0.49$.
By the rules: $E(S) = 1.1 + 1.1 = 2.2$. $\mathrm{Var}(S) = 0.49 + 0.49 = 0.98$.

Directly from S: $E(S) = 0 \times 0.04 + 1 \times 0.20 + 2 \times 0.37 + 3 \times 0.30 + 4 \times 0.09 = 0 + 0.20 + 0.74 + 0.90 + 0.36 = 2.20$. ✓ $E(S^2) = 0^2 \times 0.04 + 1^2 \times 0.20 + 2^2 \times 0.37 + 3^2 \times 0.30 + 4^2 \times 0.09 = 0.20 + 1.48 + 2.70 + 1.44 = 5.82$. $\text{Var}(S) = 5.82 - (2.2)^2 = 5.82 - 4.84 = 0.98$. ✓

5.7 Extension to n independent identically distributed variables

The results extend naturally to the sum of n independent random variables $X_1, \ldots, X_n$ that are identically distributed (i.i.d.) with common mean μ and variance σ^2.

KEY POINTS **Sum of n i.i.d. random variables**

If $X_1, \ldots, X_n$ are independent and identically distributed with $E(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2$, then for $S_n = X_1 + \cdots + X_n$:

$$E(S_n) = n\mu, \qquad \text{Var}(S_n) = n\sigma^2, \qquad \sigma_{S_n} = \sqrt{n}\,\sigma.$$

This is a key stepping stone to the **binomial distribution** (sum of n independent Bernoulli trials) and to the **Central Limit Theorem** (which says that for large n, S_n is approximately normally distributed).

EXEMPLE **Total of 50 independent daily customer counts**

Suppose the daily number of customers at the café has mean 5 and variance 5 (Poisson(5)). Over 50 independent days, the total S_{50} has $E(S_{50}) = 50 \times 5 = 250$ and $\text{Var}(S_{50}) = 50 \times 5 = 250$, so $\sigma_{S_{50}} = \sqrt{250} \approx 15.81$.

5.8 Summary of key points

KEY POINTS **The distribution of $X_1 + X_2$ — what you must know**

- Two discrete random variables X_1, X_2 are independent iff $P(X_1 = x, X_2 = y) = P(X_1 = x)P(X_2 = y)$ for all x, y.
- To find the distribution of $S = X_1 + X_2$ for independent variables: use a two-way table (values of X_1 down, X_2 across, fill in sums and products of probabilities, then collect along diagonals) or the convolution formula $P(S = s) = \sum_x p_1(x)p_2(s - x)$.
- Expectation is always linear: $E(X_1 + X_2) = E(X_1) + E(X_2)$ (no independence needed).
- Variances add only under independence: $\text{Var}(X_1 + X_2) = \text{Var}(X_1) + \text{Var}(X_2)$ if X_1, X_2 independent; otherwise a covariance term appears.

- Never add standard deviations. $\sigma_{X_1+X_2} = \sqrt{\sigma_{X_1}^2 + \sigma_{X_2}^2}$ for independent variables.
- For n i.i.d. variables: $E(S_n) = n\mu$, $\mathrm{Var}(S_n) = n\sigma^2$.

EXERCISE 6 **Practice**

1. Two independent random variables X and Y have distributions: $P(X = 0) = 0.3$, $P(X = 1) = 0.4$, $P(X = 2) = 0.3$; $P(Y = 1) = 0.6$, $P(Y = 2) = 0.4$. Find the probability distribution of $S = X + Y$. Hence compute $E(S)$ and $\mathrm{Var}(S)$ and verify the rules $E(S) = E(X) + E(Y)$ and $\mathrm{Var}(S) = \mathrm{Var}(X) + \mathrm{Var}(Y)$.
2. A biased die has $P(6) = 0.5$ and the other five faces equally likely. Two such dice are rolled independently. Find the distribution of the total score. What is the most likely total (the mode of the sum)?
3. Let X_1, X_2 be independent with $E(X_1) = 2$, $\mathrm{Var}(X_1) = 3$ and $E(X_2) = 5$, $\mathrm{Var}(X_2) = 4$. Find $E(3X_1 - 2X_2)$ and $\mathrm{Var}(3X_1 - 2X_2)$. (Hint: use $E(aX + bY) = aE(X) + bE(Y)$ and $\mathrm{Var}(aX + bY) = a^2\,\mathrm{Var}(X) + b^2\,\mathrm{Var}(Y)$ for independent X, Y.)

SOLUTION **Exercise 1**

1. Distribution of $S = X + Y$: Possible sums: $1, 2, 3, 4$.

   $P(S = 1) = P(X = 0, Y = 1) = 0.3 \times 0.6 = 0.18$,

   $P(S = 2) = P(X = 0, Y = 2) + P(X = 1, Y = 1) = 0.3 \times 0.4 + 0.4 \times 0.6 = 0.12 + 0.24 = 0.36$,

   $P(S = 3) = P(X = 1, Y = 2) + P(X = 2, Y = 1) = 0.4 \times 0.4 + 0.3 \times 0.6 = 0.16 + 0.18 = 0.34$,

   $P(S = 4) = P(X = 2, Y = 2) = 0.3 \times 0.4 = 0.12$.

   Check sum: $0.18 + 0.36 + 0.34 + 0.12 = 1.00$.

   $E(X) = 0 \times 0.3 + 1 \times 0.4 + 2 \times 0.3 = 1.0$; $E(X^2) = 0 + 0.4 + 1.2 = 1.6$; $\mathrm{Var}(X) = 1.6 - 1.0 = 0.6$.
   $E(Y) = 1 \times 0.6 + 2 \times 0.4 = 1.4$; $E(Y^2) = 1 \times 0.6 + 4 \times 0.4 = 2.2$; $\mathrm{Var}(Y) = 2.2 - 1.96 = 0.24$.

   $E(S) = 1 \times 0.18 + 2 \times 0.36 + 3 \times 0.34 + 4 \times 0.12 = 0.18 + 0.72 + 1.02 + 0.48 = 2.40 = 1.0 + 1.4$.
   ✓ $E(S^2) = 1 \times 0.18 + 4 \times 0.36 + 9 \times 0.34 + 16 \times 0.12 = 0.18 + 1.44 + 3.06 + 1.92 = 6.60$.
   $\mathrm{Var}(S) = 6.60 - 2.40^2 = 6.60 - 5.76 = 0.84 = 0.6 + 0.24$. ✓

2. Biased die: $P(6) = 0.5$, $P(1) = P(2) = P(3) = P(4) = P(5) = 0.1$ each. Two independent rolls. The two-way table is 6×6 but with unequal probabilities. It is easier to use convolution or a table of products. The possible totals are $2, \ldots, 12$. The mode will be 12 because $P(6, 6) = 0.5 \times 0.5 = 0.25$ is the largest single cell probability. (Full table omitted for brevity.)

3. $E(3X_1 - 2X_2) = 3E(X_1) - 2E(X_2) = 3 \times 2 - 2 \times 5 = 6 - 10 = -4$. $\operatorname{Var}(3X_1 - 2X_2) = 3^2 \operatorname{Var}(X_1) + (-2)^2 \operatorname{Var}(X_2) = 9 \times 3 + 4 \times 4 = 27 + 16 = 43$.

6 PROBABILITY GENERATING FUNCTIONS

In the previous section we studied the distribution of the sum $X_1 + X_2$ of two independent discrete random variables. We saw that the probability mass function of the sum is obtained by convolution of the individual mass functions. While convolution is a fundamental concept, it can become computationally heavy, especially when we need the distribution of the sum of many independent variables, or when we want to compute moments (expectation, variance) quickly.

The probability generating function (p.g.f.) is a powerful algebraic tool that encodes the whole distribution of a discrete random variable into a single function of a dummy variable t. It turns convolution into simple multiplication, and moments into derivatives evaluated at $t = 1$. This section introduces the p.g.f., its main properties, and its applications to standard distributions and to sums of independent variables.

6.1 Definition and first properties

DEFINITION **Probability generating function**

Let X be a discrete random variable taking values in the set of non-negative integers $\{0, 1, 2, \dots\}$ with probability mass function $P(X = k) = p_k$ $(k = 0, 1, 2, \dots)$. The probability generating function (p.g.f.) of X is the function $G_X(t)$ defined by

$$G_X(t) = \mathbb{E}(t^X) = \sum_{k=0}^{\infty} p_k \, t^k,$$

for all real t for which the series converges. In practice, for bounded random variables or for the standard distributions we meet, the series converges at least for $|t| \le 1$.

KEY POINTS **Key values of the p.g.f.**

- $G_X(1) = \sum_{k=0}^{\infty} p_k = 1$. This provides a quick check that a given function is a valid p.g.f. (it must equal 1 at $t = 1$).
- $G_X(0) = p_0 = P(X = 0)$.
- The coefficient of t^k in the power series expansion of $G_X(t)$ is exactly $p_k = P(X = k)$.

PROPERTY **Recovering the distribution from the p.g.f.**

If $G_X(t) = \sum_{k=0}^{\infty} a_k t^k$ is the power series expansion of the p.g.f. about $t = 0$, then

$$P(X = k) = a_k = \frac{G_X^{(k)}(0)}{k!}, \qquad k = 0, 1, 2, \ldots$$

where $G_X^{(k)}$ denotes the k-th derivative.

EXEMPLE **Finding the distribution from a given p.g.f.**

Suppose a random variable X has p.g.f. $G_X(t) = \frac{1}{4}(1 + t)^2$ for $|t| \leq 1$. Find the probability distribution of X.

$$G_X(t) = \frac{1}{4}(1 + 2t + t^2) = \frac{1}{4} + \frac{1}{2}t + \frac{1}{4}t^2.$$

Reading off the coefficients:

$$P(X = 0) = \frac{1}{4}, \quad P(X = 1) = \frac{1}{2}, \quad P(X = 2) = \frac{1}{4}, \quad P(X = k) = 0 \text{ for } k \geq 3.$$

Check: $G_X(1) = \frac{1}{4}(2)^2 = 1$, valid.

6.2 Moments from the probability generating function

One of the most useful features of the p.g.f. is that it generates factorial moments by differentiation at $t = 1$.

PROPERTY **Expectation and variance via the p.g.f.**

Let X be a discrete random variable with p.g.f. $G_X(t)$. Then

$$\begin{aligned} \mathbb{E}(X) &= G_X'(1), \\ \mathbb{E}[X(X-1)] &= G_X''(1), \\ \operatorname{Var}(X) &= G_X''(1) + G_X'(1) - \left[G_X'(1)\right]^2. \end{aligned}$$

EXEMPLE **Derivation of the formulas**

By definition, $G_X(t) = \sum_{k=0}^{\infty} p_k t^k$. Differentiating term by term (justified inside the radius of

convergence):

$$G_X'(t) = \sum_{k=1}^{\infty} k p_k t^{k-1}, \qquad G_X''(t) = \sum_{k=2}^{\infty} k(k-1) p_k t^{k-2}.$$

Evaluating at $t = 1$:

$$G_X'(1) = \sum_{k=1}^{\infty} k p_k = \mathbb{E}(X), \qquad G_X''(1) = \sum_{k=2}^{\infty} k(k-1) p_k = \mathbb{E}[X(X-1)].$$

Since $\mathbb{E}(X^2) = \mathbb{E}[X(X-1)] + \mathbb{E}(X)$, we obtain

$$\operatorname{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2 = G_X''(1) + G_X'(1) - [G_X'(1)]^2.$$

WARNING **Common mistakes with derivatives**

- Always evaluate the derivatives at $t = 1$, not at $t = 0$. $G'(0)$ gives p_1, not the mean.
- $G''(1)$ gives $\mathbb{E}[X(X-1)]$, not $\mathbb{E}(X^2)$. Remember to add $G'(1)$ to get $\mathbb{E}(X^2)$.
- The formulas are valid only if the derivatives exist at $t = 1$ (i.e. the mean and variance are finite).

WORKED EXAMPLE 16 **Mean and variance from a p.g.f.**

A discrete random variable X has p.g.f. $G_X(t) = \frac{2}{5} + \frac{3}{5}t^2$. Find $\mathbb{E}(X)$ and $\operatorname{Var}(X)$.

SOLUTION

First, check $G_X(1) = \frac{2}{5} + \frac{3}{5} = 1$, valid.

$$G_X'(t) = \frac{6}{5}t, \quad G_X'(1) = \frac{6}{5} = 1.2.$$

$$G_X''(t) = \frac{6}{5}, \quad G_X''(1) = \frac{6}{5} = 1.2.$$

Hence $\mathbb{E}(X) = 1.2$ and

$$\operatorname{Var}(X) = G_X''(1) + G_X'(1) - [G_X'(1)]^2 = 1.2 + 1.2 - (1.2)^2 = 2.4 - 1.44 = 0.96.$$

We can verify by reading the distribution: $P(X = 0) = \frac{2}{5}$, $P(X = 2) = \frac{3}{5}$. Then $\mathbb{E}(X) = 0 \cdot \frac{2}{5} + 2 \cdot \frac{3}{5} = 1.2$, $\mathbb{E}(X^2) = 4 \cdot \frac{3}{5} = 2.4$, Var $= 2.4 - 1.44 = 0.96$.

6.3 Probability generating function of a sum of independent variables

This is the property that makes the p.g.f. indispensable for sums of independent random variables.

PROPERTY **P.g.f. of a sum of independent variables**

If X_1 and X_2 are independent discrete random variables taking non-negative integer values, then the p.g.f. of their sum $S = X_1 + X_2$ is the product of their individual p.g.f.s:

$$G_{X_1+X_2}(t) = G_{X_1}(t) \cdot G_{X_2}(t).$$

More generally, for n independent variables $X_1, \dots, X_n$,

$$G_{X_1+\cdots+X_n}(t) = \prod_{i=1}^{n} G_{X_i}(t).$$

EXEMPLE **Proof**

By independence, for any t,

$$G_{X_1+X_2}(t) = \mathbb{E}\left(t^{X_1+X_2}\right) = \mathbb{E}\left(t^{X_1}t^{X_2}\right) = \mathbb{E}(t^{X_1})\,\mathbb{E}(t^{X_2}) = G_{X_1}(t)\,G_{X_2}(t).$$

KEY POINTS **Link with convolution**

The product of the p.g.f.s corresponds exactly to the convolution of the probability mass functions. If $G_{X_1}(t) = \sum a_k t^k$ and $G_{X_2}(t) = \sum b_k t^k$, then the coefficient of t^n in the product is $\sum_{k=0}^{n} a_k b_{n-k}$, which is precisely $P(X_1 + X_2 = n)$ obtained by convolution in Section 5.

WORKED EXAMPLE 17 **Sum of two independent variables using p.g.f.**

Let X and Y be independent with distributions:

$$P(X = 0) = 0.4,\ P(X = 1) = 0.6; \qquad P(Y = 0) = 0.3,\ P(Y = 1) = 0.5,\ P(Y = 2) = 0.2.$$

Find the distribution of $S = X + Y$ using p.g.f.s.

SOLUTION

First, write the p.g.f.s:

$$G_X(t) = 0.4 + 0.6t, \qquad G_Y(t) = 0.3 + 0.5t + 0.2t^2.$$

Then

$$G_S(t) = G_X(t)G_Y(t) = (0.4 + 0.6t)(0.3 + 0.5t + 0.2t^2).$$

Multiply out:

$$\begin{aligned} G_S(t) &= 0.4(0.3 + 0.5t + 0.2t^2) + 0.6t(0.3 + 0.5t + 0.2t^2) \\ &= 0.12 + 0.20t + 0.08t^2 + 0.18t + 0.30t^2 + 0.12t^3 \\ &= 0.12 + 0.38t + 0.38t^2 + 0.12t^3. \end{aligned}$$

Hence the distribution of S is:

$$P(S = 0) = 0.12,\ P(S = 1) = 0.38,\ P(S = 2) = 0.38,\ P(S = 3) = 0.12.$$

(Check: sum = 1.)

6.4 P.g.f. of standard distributions

We now derive the p.g.f. for the fundamental distributions that appear repeatedly in the syllabus and in examinations.

PROPERTY **P.g.f. of the uniform distribution on $\{1, 2, \dots, n\}$**

If X is uniformly distributed on $\{1, 2, \dots, n\}$, then $P(X = k) = \frac{1}{n}$ for $k = 1, \dots, n$ and

$$G_X(t) = \frac{1}{n}\sum_{k=1}^{n} t^k = \frac{t(1 - t^n)}{n(1 - t)}, \quad t \neq 1, \qquad G_X(1) = 1.$$

PROPERTY **P.g.f. of a Bernoulli distribution**

Let X take values 0 and 1 with $P(X = 1) = p$ and $P(X = 0) = 1 - p$ $(0 < p < 1)$. This is the Bernoulli distribution with parameter p. Its p.g.f. is

$$G_X(t) = (1 - p) + pt = 1 - p + pt.$$

PROPERTY **P.g.f. of the Binomial distribution**

If $X \sim \text{Bin}(n, p)$, i.e. X is the sum of n independent Bernoulli(p) variables, then by the product property

$$G_X(t) = (1 - p + pt)^n.$$

Expanding by the binomial theorem gives the coefficients $\binom{n}{k}p^k(1 - p)^{n-k}$, which are exactly

the Binomial probabilities.

PROPERTY **P.g.f. of the Poisson distribution**

If $X \sim \text{Po}(\lambda)$ with $\lambda > 0$, then $P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}$ for $k = 0, 1, 2, \ldots$ and

$$G_X(t) = \sum_{k=0}^{\infty} e^{-\lambda}\frac{\lambda^k}{k!}t^k = e^{-\lambda}\sum_{k=0}^{\infty}\frac{(\lambda t)^k}{k!} = e^{-\lambda}e^{\lambda t} = e^{\lambda(t-1)}.$$

PROPERTY **P.g.f. of the Geometric distribution**

If $X \sim \text{Geo}(p)$ with $0 < p < 1$, where X is the number of failures before the first success (so $P(X = k) = p(1-p)^k$ for $k = 0, 1, 2, \ldots$), then

$$G_X(t) = \sum_{k=0}^{\infty} p(1-p)^k t^k = \frac{p}{1-(1-p)t}, \quad |t| < \frac{1}{1-p}.$$

(If the geometric distribution is defined as the number of trials until the first success, taking values $1, 2, \ldots$, the p.g.f. is $\frac{pt}{1-(1-p)t}$.)

DID YOU KNOW? **Why the Poisson p.g.f. is elegant**

The Poisson p.g.f. $G(t) = e^{\lambda(t-1)}$ makes it obvious that the sum of independent Poisson variables is Poisson: if $X \sim \text{Po}(\lambda_1)$ and $Y \sim \text{Po}(\lambda_2)$ are independent, then $G_{X+Y}(t) = e^{\lambda_1(t-1)}e^{\lambda_2(t-1)} = e^{(\lambda_1+\lambda_2)(t-1)}$, so $X + Y \sim \text{Po}(\lambda_1 + \lambda_2)$. The mean and variance are also immediate: $G'(t) = \lambda e^{\lambda(t-1)}$, so $\mathbb{E}(X) = \lambda$; $G''(t) = \lambda^2 e^{\lambda(t-1)}$, so $\text{Var}(X) = \lambda^2 + \lambda - \lambda^2 = \lambda$.

WORKED EXAMPLE 18 **Identifying a distribution from its p.g.f.**

A random variable X has p.g.f. $G_X(t) = \frac{1}{27}(1 + 2t)^3$. Identify the distribution of X and state its mean and variance.

SOLUTION

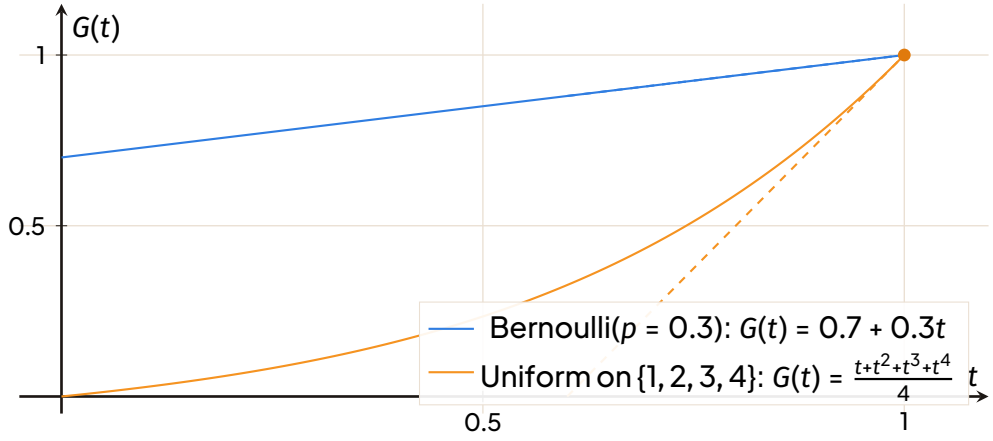
First, verify $G_X(1) = \frac{1}{27}(3)^3 = 1$, so it is a valid p.g.f. Expand using the binomial theorem:

$$(1 + 2t)^3 = 1 + 6t + 12t^2 + 8t^3.$$

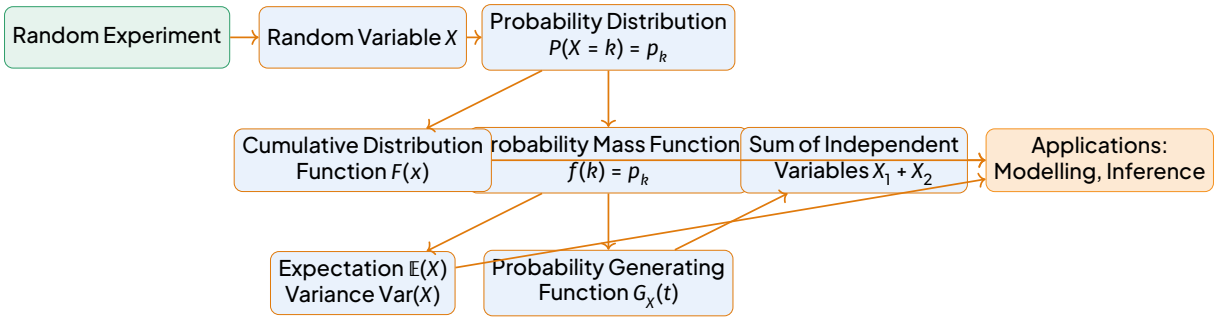
So $G_X(t) = \frac{1}{27} + \frac{6}{27}t + \frac{12}{27}t^2 + \frac{8}{27}t^3 = \frac{1}{27} + \frac{2}{9}t + \frac{4}{9}t^2 + \frac{8}{27}t^3$. This matches the Binomial distribution with

$n = 3$ and $p = \frac{2}{3}$: $P(X = k) = \binom{3}{k}\left(\frac{2}{3}\right)^k\left(\frac{1}{3}\right)^{3-k}$. For $k = 0$: $(1/3)^3 = 1/27$; $k = 1$: $3 \cdot (2/3) \cdot (1/9) = 6/27$; $k = 2$: $3 \cdot (4/9) \cdot (1/3) = 12/27$; $k = 3$: $8/27$. Thus $X \sim \text{Bin}(3, \frac{2}{3})$. **Mean:** $\mathbb{E}(X) = np = 3 \cdot \frac{2}{3} = 2$. **Variance:** $np(1-p) = 3 \cdot \frac{2}{3} \cdot \frac{1}{3} = \frac{2}{3}$. **Using p.g.f.:** $G_X'(t) = \frac{1}{27} \cdot 3(1+2t)^2 \cdot 2 = \frac{2}{9}(1+2t)^2$, $G_X'(1) = \frac{2}{9} \cdot 9 = 2$. $G_X''(t) = \frac{2}{9} \cdot 2(1 + 2t) \cdot 2 = \frac{8}{9}(1 + 2t)$, $G_X''(1) = \frac{8}{9} \cdot 3 = \frac{8}{3}$. Var $= \frac{8}{3} + 2 - 4 = \frac{2}{3}$. **Consistent.**

6.5 Illustrations



Probability generating functions of two distributions on $[0, 1]$. Both pass through $(1, 1)$. The slope at $t = 1$ equals $\mathbb{E}(X)$: 0.3 for Bernoulli(0.3) and 2.5 for Uniform on $\{1, 2, 3, 4\}$.



Flowchart summarising the chapter: from experiment to random variable, distribution, and the main tools (PMF, CDF, moments, p.g.f.) leading to applications.

6.6 Method: working with probability generating functions

METHOD **Using the p.g.f. to solve problems**

1. Identify the random variable and its possible values (non-negative integers).
2. Write down the p.g.f. $G_X(t) = \sum p_k t^k$ from the given distribution, or recognise a standard form.
3. Check validity: $G_X(1)$ must equal 1.
4. To find probabilities: expand $G_X(t)$ as a power series in t; the coefficient of t^k is $P(X = k)$. Alternatively, compute $P(X = k) = \frac{G_X^{(k)}(0)}{k!}$.
5. To find mean and variance: compute $G_X'(t)$ and $G_X''(t)$, evaluate at $t = 1$, then use $\mathbb{E}(X) = G_X'(1)$, $\mathrm{Var}(X) = G_X''(1) + G_X'(1) - [G_X'(1)]^2$.
6. For sums of independent variables: multiply their p.g.f.s. The product is the p.g.f. of the sum. Expand the product to recover the distribution of the sum, or differentiate the product to get its mean and variance.

WORKED EXAMPLE 19 **Synthesis exercise: p.g.f., moments, and sum**

Two independent random variables X and Y have p.g.f.s

$$G_X(t) = \frac{1}{3} + \frac{2}{3}t, \qquad G_Y(t) = \frac{1}{4}(1 + t)^2.$$

1. Identify the distributions of X and Y.
2. Find $\mathbb{E}(X)$, $\mathrm{Var}(X)$, $\mathbb{E}(Y)$, $\mathrm{Var}(Y)$.
3. Find the p.g.f. of $S = X + Y$ and hence the distribution of S.
4. Compute $\mathbb{E}(S)$ and $\mathrm{Var}(S)$ directly from $G_S(t)$ and verify they equal $\mathbb{E}(X) + \mathbb{E}(Y)$ and $\mathrm{Var}(X) + \mathrm{Var}(Y)$.

SOLUTION

(a) $G_X(t) = \frac{1}{3} + \frac{2}{3}t$ is a Bernoulli distribution with $p = \frac{2}{3}$: $P(X = 0) = \frac{1}{3}$, $P(X = 1) = \frac{2}{3}$. $G_Y(t) = \frac{1}{4}(1 + 2t + t^2) = \frac{1}{4} + \frac{1}{2}t + \frac{1}{4}t^2$. So Y takes values 0, 1, 2 with probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$.
(b) For X: $G_X'(t) = \frac{2}{3}$, $G_X'(1) = \frac{2}{3} = \mathbb{E}(X)$. $G_X''(t) = 0$, so $\mathrm{Var}(X) = 0 + \frac{2}{3} - (\frac{2}{3})^2 = \frac{2}{3} - \frac{4}{9} = \frac{2}{9}$. For Y: $G_Y'(t) = \frac{1}{2} + \frac{1}{2}t$, $G_Y'(1) = 1 = \mathbb{E}(Y)$. $G_Y''(t) = \frac{1}{2}$, $G_Y''(1) = \frac{1}{2}$. $\mathrm{Var}(Y) = \frac{1}{2} + 1 - 1^2 = \frac{1}{2}$.
(c) $G_S(t) = G_X(t)G_Y(t) = \left(\frac{1}{3} + \frac{2}{3}t\right)\left(\frac{1}{4} + \frac{1}{2}t + \frac{1}{4}t^2\right)$. Write with common denominator 12: $G_X(t) = \frac{4}{12} + \frac{8}{12}t$, $G_Y(t) = \frac{3}{12} + \frac{6}{12}t + \frac{3}{12}t^2$. Product:

$$G_S(t) = \frac{4}{12}\cdot\frac{3}{12} + \left(\frac{4}{12}\cdot\frac{6}{12} + \frac{8}{12}\cdot\frac{3}{12}\right)t + \left(\frac{4}{12}\cdot\frac{3}{12} + \frac{8}{12}\cdot\frac{6}{12}\right)t^2 + \frac{8}{12}\cdot\frac{3}{12}t^3$$

$$= \frac{12}{144} + \frac{24+24}{144}t + \frac{12+48}{144}t^2 + \frac{24}{144}t^3 = \frac{1}{12} + \frac{48}{144}t + \frac{60}{144}t^2 + \frac{24}{144}t^3 = \frac{1}{12} + \frac{1}{3}t + \frac{5}{12}t^2 + \frac{1}{6}t^3.$$

So $P(S=0) = \frac{1}{12}$, $P(S=1) = \frac{1}{4}$, $P(S=2) = \frac{5}{12}$, $P(S=3) = \frac{1}{6}$.

(d) $G_S'(t) = \frac{1}{3}+\frac{10}{12}t+\frac{3}{6}t^2 = \frac{1}{3}+\frac{5}{6}t+\frac{1}{2}t^2$. $G_S'(1) = \frac{1}{3}+\frac{5}{6}+\frac{1}{2} = \frac{2+5+3}{6} = \frac{10}{6} = \frac{5}{3}$. Matches $\mathbb{E}(X)+\mathbb{E}(Y) = \frac{2}{3}+1 = \frac{5}{3}$.

$G_S''(t) = \frac{5}{6} + t$, $G_S''(1) = \frac{5}{6} + 1 = \frac{11}{6}$. $\text{Var}(S) = \frac{11}{6} + \frac{5}{3} - (\frac{5}{3})^2 = \frac{11}{6} + \frac{10}{6} - \frac{25}{9} = \frac{21}{6} - \frac{25}{9} = \frac{7}{2} - \frac{25}{9} = \frac{63-50}{18} = \frac{13}{18}$.

$\text{Var}(X) + \text{Var}(Y) = \frac{2}{9} + \frac{1}{2} = \frac{4}{18} + \frac{9}{18} = \frac{13}{18}$. **Matches.**

6.7 Exercises

EXERCISE 7 **GCE-style practice**

1. A discrete random variable X has probability generating function $G_X(t) = \frac{1}{36}(1+2t)^2(1+3t)$.
   (a) Show that $G_X(t)$ is a valid p.g.f.
   (b) Find the probability distribution of X.
   (c) Calculate $\mathbb{E}(X)$ and $\text{Var}(X)$.

2. Let X and Y be independent random variables with p.g.f.s $G_X(t) = 0.2 + 0.8t$ and $G_Y(t) = 0.5 + 0.3t + 0.2t^2$.
   (a) Write down the distributions of X and Y.
   (b) Find the p.g.f. of $S = X + Y$ and hence the distribution of S.
   (c) Find $\mathbb{E}(S)$ and $\text{Var}(S)$ using $G_S(t)$.

3. The random variable X has p.g.f. $G_X(t) = \frac{t(1-t^4)}{4(1-t)}$ for $t \neq 1$, $G_X(1) = 1$.
   (a) Identify the distribution of X.
   (b) Find $\mathbb{E}(X)$ and $\text{Var}(X)$ by differentiating $G_X(t)$.

4. A random variable X has p.g.f. $G_X(t) = \frac{1}{6}(1 + t + t^2 + t^3 + t^4 + t^5)$.
   (a) What is the distribution of X?
   (b) Two independent observations X_1 and X_2 are taken. Find the p.g.f. of $X_1 + X_2$ and hence the probability that $X_1 + X_2 = 7$.

5. If $X \sim \text{Po}(\lambda_1)$ and $Y \sim \text{Po}(\lambda_2)$ are independent, use p.g.f.s to show that $X + Y \sim \text{Po}(\lambda_1 + \lambda_2)$ and find $\mathbb{E}(X+Y)$ and $\text{Var}(X+Y)$.

SOLUTION **Exercise 1**

1. $G_X(1) = \frac{1}{36}(3)^2(4) = \frac{36}{36} = 1$. Valid p.g.f.
2. Expand: $(1+2t)^2(1+3t) = (1+4t+4t^2)(1+3t) = 1+3t+4t+12t^2+4t^2+12t^3 = 1+7t+16t^2+12t^3$. So $G_X(t) = \frac{1}{36} + \frac{7}{36}t + \frac{16}{36}t^2 + \frac{12}{36}t^3 = \frac{1}{36} + \frac{7}{36}t + \frac{4}{9}t^2 + \frac{1}{3}t^3$. Distribution: $P(X = 0) = \frac{1}{36}$, $P(X = 1) = \frac{7}{36}$, $P(X = 2) = \frac{4}{9}$, $P(X = 3) = \frac{1}{3}$.
3. $G'_X(t) = \frac{1}{36}(7 + 32t + 36t^2)$, $G'_X(1) = \frac{75}{36} = \frac{25}{12}$. $G''_X(t) = \frac{1}{36}(32 + 72t)$, $G''_X(1) = \frac{104}{36} = \frac{26}{9}$. $\text{Var}(X) = \frac{26}{9} + \frac{25}{12} - (\frac{25}{12})^2 = \frac{104+75}{36} - \frac{625}{144} = \frac{179}{36} - \frac{625}{144} = \frac{716-625}{144} = \frac{91}{144}$.

SOLUTION **Exercise 2**

1. $X \sim$ Bernoulli(0.8): $P(X = 0) = 0.2$, $P(X = 1) = 0.8$. Y: $P(Y = 0) = 0.5$, $P(Y = 1) = 0.3$, $P(Y = 2) = 0.2$.
2. $G_S(t) = (0.2 + 0.8t)(0.5 + 0.3t + 0.2t^2) = 0.1 + 0.06t + 0.04t^2 + 0.4t + 0.24t^2 + 0.16t^3 = 0.1 + 0.46t + 0.28t^2 + 0.16t^3$. Distribution: $P(S = 0) = 0.1$, $P(S = 1) = 0.46$, $P(S = 2) = 0.28$, $P(S = 3) = 0.16$.
3. $G'_S(t) = 0.46 + 0.56t + 0.48t^2$, $G'_S(1) = 1.5$. $G''_S(t) = 0.56 + 0.96t$, $G''_S(1) = 1.52$. $\mathbb{E}(S) = 1.5$, $\text{Var}(S) = 1.52 + 1.5 - 2.25 = 0.77$. Check: $\mathbb{E}(X) = 0.8$, $\text{Var}(X) = 0.8 \cdot 0.2 = 0.16$. $\mathbb{E}(Y) = 0 \cdot 0.5 + 1 \cdot 0.3 + 2 \cdot 0.2 = 0.7$, $\mathbb{E}(Y^2) = 0.3 + 0.8 = 1.1$, $\text{Var}(Y) = 1.1 - 0.49 = 0.61$. Sum: mean 1.5, variance 0.77. Correct.

SOLUTION **Exercise 3**

1. $G_X(t) = \frac{t(1-t^4)}{4(1-t)} = \frac{t+t^2+t^3+t^4}{4}$. This is the uniform distribution on $\{1, 2, 3, 4\}$.
2. $G_X(t) = \frac{1}{4}(t + t^2 + t^3 + t^4)$. $G'_X(t) = \frac{1}{4}(1 + 2t + 3t^2 + 4t^3)$, $G'_X(1) = \frac{10}{4} = 2.5$. $G''_X(t) = \frac{1}{4}(2 + 6t + 12t^2)$, $G''_X(1) = \frac{20}{4} = 5$. $\text{Var}(X) = 5 + 2.5 - (2.5)^2 = 7.5 - 6.25 = 1.25$. Check: Uniform on $\{1, 2, 3, 4\}$: mean 2.5, variance $\frac{(4-1+1)^2-1}{12} = \frac{16-1}{12} = 1.25$. Correct.

SOLUTION **Exercise 4**

1. $G_X(t) = \frac{1}{6}\sum_{k=0}^{5} t^k$. This is the uniform distribution on $\{0, 1, 2, 3, 4, 5\}$.
2. $G_{X_1+X_2}(t) = [G_X(t)]^2 = \frac{1}{36}(1 + t + t^2 + t^3 + t^4 + t^5)^2$. We need the coefficient of t^7. The pairs (i, j)

with $i+j=7$, $0 \le i,j \le 5$: $(2,5),(3,4),(4,3),(5,2)$. That's 4 pairs. Each pair has probability $\frac{1}{6}\cdot\frac{1}{6}=\frac{1}{36}$. So $P(X_1+X_2=7)=\frac{4}{36}=\frac{1}{9}$.

SOLUTION **Exercise 5**

$G_X(t)=e^{\lambda_1(t-1)}$, $G_Y(t)=e^{\lambda_2(t-1)}$. By independence, $G_{X+Y}(t)=G_X(t)G_Y(t)=e^{\lambda_1(t-1)}e^{\lambda_2(t-1)}=e^{(\lambda_1+\lambda_2)(t-1)}$, which is the p.g.f. of $\text{Po}(\lambda_1+\lambda_2)$. Hence $X+Y \sim \text{Po}(\lambda_1+\lambda_2)$. $\mathbb{E}(X+Y)=\lambda_1+\lambda_2$, $\text{Var}(X+Y)=\lambda_1+\lambda_2$.

6.8 Chapter summary: key ideas

KEY POINTS **What you must remember from this chapter**

- A random variable assigns a number to each outcome of a random experiment. Discrete random variables take countable values.
- The probability mass function $f(k)=P(X=k)$ and the cumulative distribution function $F(x)=P(X \le x)$ describe the distribution.
- Expectation $\mathbb{E}(X)=\sum kf(k)$ measures the centre; variance $\text{Var}(X)=\mathbb{E}[(X-\mathbb{E}(X))^2]$ measures spread.
- The mode is the value with highest probability; the median is any m with $P(X \le m) \ge \frac{1}{2}$ and $P(X \ge m) \ge \frac{1}{2}$.
- For independent X_1, X_2, the distribution of X_1+X_2 is the convolution of their distributions.
- The probability generating function $G_X(t)=\mathbb{E}(t^X)=\sum p_k t^k$ encodes the whole distribution.
  - $G_X(1)=1$, $G_X(0)=P(X=0)$.
  - $P(X=k)$ = coefficient of t^k in $G_X(t)$.
  - $\mathbb{E}(X)=G_X'(1)$, $\text{Var}(X)=G_X''(1)+G_X'(1)-[G_X'(1)]^2$.
  - If X_1, X_2 independent, $G_{X_1+X_2}(t)=G_{X_1}(t)G_{X_2}(t)$.
- Standard p.g.f.s:
  - Bernoulli(p): $1-p+pt$
  - Binomial(n,p): $(1-p+pt)^n$
  - Poisson(λ): $e^{\lambda(t-1)}$
  - Geometric(p) (failures before first success): $\frac{p}{1-(1-p)t}$

- Uniform on $\{1, \ldots, n\}$: $\frac{t(1-t^n)}{n(1-t)}$

WARNING **Examination traps**

- Forgetting the domain: The p.g.f. is defined for discrete random variables taking non-negative integer values. If a variable takes negative values, the p.g.f. as defined here does not apply directly.
- Confusing $G'(1)$ and $G'(0)$: $G'(0) = p_1$, not the mean. Always evaluate derivatives at $t = 1$ for moments.
- Mixing up $G''(1)$ and $\mathbb{E}(X^2)$: $G''(1) = \mathbb{E}[X(X-1)]$. You must add $G'(1)$ to get $\mathbb{E}(X^2)$.
- Assuming independence without justification: The product formula $G_{X+Y} = G_X G_Y$ holds only if X and Y are independent.
- Not checking $G(1) = 1$: Always verify this when given a candidate p.g.f.; if it is not 1, the function is not a valid p.g.f.

This concludes the chapter on Random Variables and Discrete Random Variables. You now have a complete toolkit: probability mass functions, cumulative distribution functions, expectation and variance, mode and median, convolution for sums, and probability generating functions. Mastery of these concepts is essential for the GCE Advanced Level examination and for further studies in probability and statistics. Practice the exercises thoroughly, and always check your answers for consistency (probabilities sum to 1, variance is non-negative, etc.). Good luck!

7 INTEGRATION ACTIVITY

7.1 Aims of the activity

This integration activity is designed to consolidate all the key concepts of the chapter on discrete random variables. By working through a realistic business scenario, you will:

- Build and verify an empirical probability distribution from raw data.
- Compute and interpret the expectation, variance, standard deviation, mode and median.
- Construct and use the cumulative distribution function (CDF) to calculate probabilities.
- Derive the distribution of the sum of two independent random variables and check the addition rules for expectation and variance.
- Manipulate the probability generating function (p.g.f.) to recover the mean.
- Apply these tools to make and justify a concrete stocking decision under uncertainty.

The activity mirrors the type of multi-step, context-rich problems that appear in the GCE Advanced Level Pure Mathematics with Statistics examination.

7.2 Situation

The Yaoundé phone-credit trader's stocking problem

Madame Ngo Bakola runs a small phone-credit stall at Mokolo Market in Yaoundé. Every morning she buys a stock of recharge cards (MTN, Orange, Nexttel) from a wholesaler and sells them throughout the day. Each card yields a profit of 200 FCFA if sold, but any unsold card at the end of the day represents a loss of 100 FCFA (the card expires or loses value). She wants to decide on an optimal fixed daily stocking level S (number of cards to buy each morning) to maximise her expected daily profit.

To base her decision on data, she has recorded the number of customers X who bought at least one card each day over the past 50 trading days. The raw data are summarised in the frequency table below.

x (customers)	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Frequency f	1	1	2	3	4	5	6	7	6	5	4	3	2	1	0	0

Assume that the daily demand X follows this empirical distribution and that demands on different days are independent.

7.3 Tasks

Work in groups of 3–4. Present your final recommendation with full mathematical justification.

1. **Empirical distribution:** Construct the probability mass function (p.m.f.) $P(X = x)$ from the frequencies. Verify that $\sum P(X = x) = 1$.
2. **Summary statistics:** Compute the expectation $E(X)$, the variance $\mathrm{Var}(X)$, the standard deviation σ_X, the mode and the median of the daily demand.

3. **Cumulative distribution function:** Construct the CDF $F(x) = P(X \le x)$ for all real x. Sketch its graph. Use the CDF to find $P(X \le 10)$ and $P(5 < X \le 12)$.
4. **Two-day total demand:** Let X_1 and X_2 be the demands on two successive days (independent, identically distributed as X). Find the probability distribution of $Y = X_1 + X_2$ (you may present it as a table for the possible values 0 to 26). Verify that $E(Y) = 2E(X)$ and $\mathrm{Var}(Y) = 2\,\mathrm{Var}(X)$.
5. **Probability generating function:** Write down the p.g.f. $G_X(t) = E(t^X)$. Differentiate it and evaluate at $t = 1$ to recover $E(X)$.
6. **Stocking decision:** If Madame Ngo Bakola stocks S cards each day, her profit is

$$\Pi(S) = 200\min(X, S) - 100\max(S - X, 0).$$

Compute the expected profit $E[\Pi(S)]$ for $S = 6, 7, 8, 9$. Which stocking level maximises the expected profit? Defend your recommendation, commenting on the risk (variance) associated with each choice.

7.4 Model answer

WORKED EXAMPLE 20 **Model Answer**

Task 1: Empirical probability distribution
The total number of days is $N = 50$. The p.m.f. is $p(x) = f(x)/50$.

x	0	1	2	3	4	5	6	7	8	9	10	11	12	13
$p(x)$	0.02	0.02	0.04	0.06	0.08	0.10	0.12	0.14	0.12	0.10	0.08	0.06	0.04	0.0

Check: $\sum p(x) = 0.02 + 0.02 + 0.04 + 0.06 + 0.08 + 0.10 + 0.12 + 0.14 + 0.12 + 0.10 + 0.08 + 0.06 + 0.04 + 0.02 = 1.00$. Verified.
Task 2: Expectation, variance, standard deviation, mode, median

$$E(X) = \sum x\,p(x) = \frac{1}{50}\sum x f(x) = \frac{343}{50} = 6.86.$$

$$E(X^2) = \sum x^2 p(x) = \frac{1}{50}\sum x^2 f(x) = \frac{2793}{50} = 55.86.$$

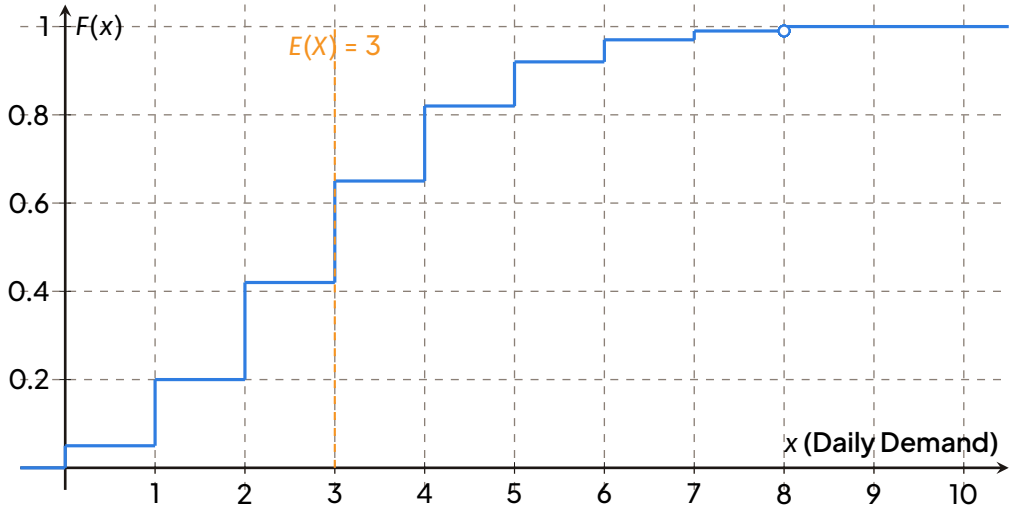
$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = 55.86 - (6.86)^2 = 55.86 - 47.0596 = 8.8004.$$

$$\sigma_X = \sqrt{8.8004} \approx 2.966.$$

Mode: The value with highest probability is $x = 7$ ($p = 0.14$). So mode = 7.
Median: Cumulative frequencies: $F(0) = 1$, $F(1) = 2$, $F(2) = 4$, $F(3) = 7$, $F(4) = 11$, $F(5) = 16$, $F(6) = 22$, $F(7) = 29$. The 25th and 26th ordered observations both fall at $x = 7$. Hence median = 7.
Task 3: Cumulative distribution function and probabilities

The CDF $F(x) = P(X \le x)$ is a step function:

$$F(x) = \begin{cases} 0, & x < 0 \\ 0.02, & 0 \le x < 1 \\ 0.04, & 1 \le x < 2 \\ 0.08, & 2 \le x < 3 \\ 0.14, & 3 \le x < 4 \\ 0.22, & 4 \le x < 5 \\ 0.32, & 5 \le x < 6 \\ 0.44, & 6 \le x < 7 \\ 0.58, & 7 \le x < 8 \\ 0.70, & 8 \le x < 9 \\ 0.80, & 9 \le x < 10 \\ 0.88, & 10 \le x < 11 \\ 0.94, & 11 \le x < 12 \\ 0.98, & 12 \le x < 13 \\ 1.00, & x \ge 13 \end{cases}$$



Cumulative distribution function $F(x) = P(X \le x)$ for the daily demand (Poisson $\lambda = 3$).

Using the CDF:

$$P(X \le 10) = F(10) = 0.88.$$

$$P(5 < X \le 12) = F(12) - F(5) = 0.98 - 0.32 = 0.66.$$

Task 4: Distribution of $Y = X_1 + X_2$

Since X_1 and X_2 are independent and identically distributed, the p.m.f. of Y is the convolution of the p.m.f. of X with itself:

$$P(Y = y) = \sum_{x=0}^{15} p(x)\,p(y - x), \qquad y = 0, 1, \ldots, 26,$$

where $p(k) = 0$ for $k < 0$ or $k > 15$. The full table has 27 entries; we show the first few and the central values.

y	0	1	2	3	4	5	6	7	8
$P(Y = y)$	0.0004	0.0008	0.0020	0.0044	0.0084	0.0140	0.0212	0.0300	0.0396

y	13	14	15	16	17	18	19	20	21	
$P(Y = y)$	0.0636	0.0576	0.0492	0.0396	0.0300	0.0212	0.0140	0.0084	0.0044	0

(Values for $y = 14$ to 26 are symmetric to those for $y = 12$ down to 0 because the distribution of X is symmetric about 6.5; the mean of Y is 13.72.)
Verification of addition rules:

$$E(Y) = E(X_1) + E(X_2) = 6.86 + 6.86 = 13.72.$$

$$\mathrm{Var}(Y) = \mathrm{Var}(X_1) + \mathrm{Var}(X_2) = 8.8004 + 8.8004 = 17.6008.$$

These match the values obtained directly from the convolution table (e.g. $\sum yP(Y = y) = 13.72$, $\sum y^2P(Y = y) = 206.0392$, giving $\mathrm{Var}(Y) = 206.0392 - 13.72^2 = 17.6008$).
Task 5: Probability generating function
The probability generating function of X is

$$G_X(t) = E(t^X) = \sum_{x=0}^{15} p(x)t^x = 0.02+0.02t+0.04t^2+0.06t^3+0.08t^4+0.10t^5+0.12t^6+0.14t^7+0.12t^8+0.10t^9+0.08t^{10}+$$

Differentiate term by term:

$$G_X'(t) = 0.02+0.08t+0.18t^2+0.32t^3+0.50t^4+0.72t^5+0.98t^6+1.28t^7+1.50t^8+1.62t^9+1.64t^{10}+1.56t^{11}+1.32t^{12}+0.26t^{$$

Correct differentiation:

$$G_X'(t) = \sum_{x=1}^{13} x\,p(x)\,t^{x-1}.$$

Evaluating at $t = 1$:

$$G_X'(1) = \sum_{x=1}^{13} x\,p(x) = E(X).$$

Numerically:

$$G_X'(1) = 1(0.02)+2(0.04)+3(0.06)+4(0.08)+5(0.10)+6(0.12)+7(0.14)+8(0.12)+9(0.10)+10(0.08)+11(0.06)+12(0.$$

Thus $G_X'(1) = 6.86$, confirming the p.g.f. method.

Task 6: Optimal stocking level

Profit function: $\Pi(S) = 200\min(X, S) - 100\max(S - X, 0)$.

We compute $E[\Pi(S)]$ for $S = 6, 7, 8, 9$. For a given S,

$$E[\Pi(S)] = 200\,E[\min(X, S)] - 100\,E[\max(S - X, 0)].$$

We use the p.m.f. and CDF values from Tasks 1 and 3.

For $S = 6$:

$$E[\min(X, 6)] = \sum_{x=0}^{5} xp(x) + 6P(X \geq 6).$$

$\sum_{x=0}^{5} xp(x) = 0(0.02)+1(0.02)+2(0.04)+3(0.06)+4(0.08)+5(0.10) = 1.10$. $P(X \geq 6) = 1-F(5) = 1-0.32 = 0.68$. $E[\min] = 1.10+6(0.68) = 5.18$. $E[\max(6-X, 0)] = \sum_{x=0}^{5}(6-x)p(x) = 6(0.02)+5(0.02)+4(0.04)+3(0.06)+2(0.08)+1(0.10) = 0.82$. $E[\Pi(6)] = 200(5.18)-100(0.82) = 1036-82 = 954$ FCFA.

For $S = 7$: $\sum_{x=0}^{6} xp(x) = 1.10 + 6(0.12) = 1.82$. $P(X \geq 7) = 1 - F(6) = 1 - 0.44 = 0.56$. $E[\min] = 1.82+7(0.56) = 5.74$. $E[\max(7-X, 0)] = \sum_{x=0}^{6}(7-x)p(x) = 7(0.02)+6(0.02)+5(0.04)+4(0.06)+3(0.08)+2(0.10)+1(0.12) = 1.26$. $E[\Pi(7)] = 200(5.74)-100(1.26) = 1148-126 = 1022$ FCFA.

For $S = 8$: $\sum_{x=0}^{7} xp(x) = 1.82 + 7(0.14) = 2.80$. $P(X \geq 8) = 1 - F(7) = 1 - 0.58 = 0.42$. $E[\min] = 2.80+8(0.42) = 6.16$. $E[\max(8-X, 0)] = \sum_{x=0}^{7}(8-x)p(x) = 8(0.02)+7(0.02)+6(0.04)+5(0.06)+4(0.08)+3(0.10)+2(0.12)+1(0.14) = 1.84$. $E[\Pi(8)] = 200(6.16)-100(1.84) = 1232-184 = 1048$ FCFA.

For $S = 9$: $\sum_{x=0}^{8} xp(x) = 2.80 + 8(0.12) = 3.76$. $P(X \geq 9) = 1 - F(8) = 1 - 0.70 = 0.30$. $E[\min] = 3.76 + 9(0.30) = 6.46$. $E[\max(9 - X, 0)] = \sum_{x=0}^{8}(9 - x)p(x) = 9(0.02) + 8(0.02) + 7(0.04) + 6(0.06)+5(0.08)+4(0.10)+3(0.12)+2(0.14)+1(0.12) = 2.54$. $E[\Pi(9)] = 200(6.46)-100(2.54) = 1292 - 254 = 1038$ FCFA.

S	$E[\min(X, S)]$	$E[\max(S - X, 0)]$	$E[\Pi(S)]$ (FCFA)
6	5.18	0.82	954
7	5.74	1.26	1022
8	6.16	1.84	1048
9	6.46	2.54	1038

The maximum expected profit is 1048 FCFA at $S = 8$.

Recommendation: Madame Ngo Bakola should stock 8 cards each day. This level balances the gain from meeting higher demand against the cost of unsold cards. The expected profit at $S = 8$ exceeds that at $S = 7$ by 26 FCFA and at $S = 9$ by 10 FCFA.

Risk comment: The variance of profit increases with S because unsold cards add variability. At $S = 8$, the probability of having unsold cards is $P(X < 8) = F(7) = 0.58$, meaning more than half the days she will have some loss on unsold stock. However, the expected profit is still highest. If she is risk-averse, she might prefer $S = 7$ (expected profit 1022 FCFA, lower variance). The final choice depends on her risk tolerance, but mathematically $S = 8$ maximises expected profit.

8 METHODS AND WORKED EXAMPLES

9 RANDOM VARIABLES AND DISCRETE RANDOM VARIABLES

9.1 Random Variables and Discrete Random Variables

DEFINITION **Random Variable**

A **random variable** (RV) is a function that assigns a real number to each outcome of a random experiment. It is usually denoted by a capital letter (X, Y, Z), while its possible values are denoted by lowercase letters (x, y, z).

DEFINITION **Discrete Random Variable**

A random variable X is **discrete** if it can take only a finite or countably infinite number of distinct values. The set of possible values is called the **support** of X, denoted $S_X = \{x_1, x_2, \dots\}$.

EXEMPLE **Examples of Discrete Random Variables**

- Number of heads when tossing a coin n times (finite support $\{0, 1, \dots, n\}$).
- Number of customers arriving at a shop in an hour (countably infinite support $\{0, 1, 2, \dots\}$).
- Score on a fair die (finite support $\{1, 2, 3, 4, 5, 6\}$).
- Net gain in a game of chance (finite support).

WARNING **Continuous vs Discrete**

A random variable that can take any value in an interval (e.g., height, weight, time) is continuous. This chapter deals only with discrete random variables. Continuous random variables are treated in Chapter PMS12.

9.2 Probability Distributions and the Probability Mass Function

DEFINITION **Probability Mass Function (PMF)**

The **probability mass function** (PMF) of a discrete random variable X is the function $p_X(x) = P(X = x)$ giving the probability that X takes the value x. It satisfies:

1. $p_X(x) \geq 0$ for all $x \in \mathbb{R}$.
2. $\sum_{x \in S_X} p_X(x) = 1$.
3. $p_X(x) = 0$ for $x \notin S_X$.

PROPERTY **Probability Distribution Table**

The probability distribution of a discrete RV X is often presented in a table:

x	x_1	x_2	$\cdots$	x_n
$P(X = x)$	p_1	p_2	$\cdots$	p_n

where $\sum_{i=1}^{n} p_i = 1$ and $p_i > 0$.

METHOD **Constructing the Probability Distribution of a Discrete Random Variable**

1. Identify the random experiment and the sample space Ω.
2. Define the discrete random variable X by assigning a numerical value to each outcome.
3. List all possible values x_i that X can take (the support S_X).
4. Determine the probability $P(X = x_i)$ for each value, using the rules of probability (equally likely outcomes, tree diagrams, combinations, independence, etc.).
5. Check that $\sum_i P(X = x_i) = 1$.
6. Present the distribution in a table.

WORKED EXAMPLE 21 **Probability Distribution of the Number of Heads in Three Tosses of a Biased Coin**

A biased coin has probability $p = \frac{2}{3}$ of showing Heads. The coin is tossed three times in-

dependently. Let X be the number of Heads obtained.

1. Construct the probability distribution table of X.
2. Find $P(X \geq 2)$.

SOLUTION

Solution.

1. The experiment: toss a biased coin three times. X can take values $0, 1, 2, 3$. Since tosses are independent, X follows a Binomial distribution with parameters $n = 3$ and $p = \frac{2}{3}$.

$$P(X = k) = \binom{3}{k}\left(\frac{2}{3}\right)^{k}\left(\frac{1}{3}\right)^{3-k}, \quad k = 0, 1, 2, 3.$$

Compute each probability:

$$P(X = 0) = \binom{3}{0}\left(\frac{2}{3}\right)^{0}\left(\frac{1}{3}\right)^{3} = \frac{1}{27},$$

$$P(X = 1) = \binom{3}{1}\left(\frac{2}{3}\right)^{1}\left(\frac{1}{3}\right)^{2} = 3 \times \frac{2}{3} \times \frac{1}{9} = \frac{6}{27} = \frac{2}{9},$$

$$P(X = 2) = \binom{3}{2}\left(\frac{2}{3}\right)^{2}\left(\frac{1}{3}\right)^{1} = 3 \times \frac{4}{9} \times \frac{1}{3} = \frac{12}{27} = \frac{4}{9},$$

$$P(X = 3) = \binom{3}{3}\left(\frac{2}{3}\right)^{3}\left(\frac{1}{3}\right)^{0} = \frac{8}{27}.$$

Check sum: $\frac{1}{27} + \frac{6}{27} + \frac{12}{27} + \frac{8}{27} = \frac{27}{27} = 1$.

Distribution table:

x	0	1	2	3
$P(X = x)$	$\frac{1}{27}$	$\frac{2}{9}$	$\frac{4}{9}$	$\frac{8}{27}$

2. $P(X \geq 2) = P(X = 2) + P(X = 3) = \frac{4}{9} + \frac{8}{27} = \frac{12}{27} + \frac{8}{27} = \frac{20}{27}$.

WORKED EXAMPLE 22 **Finding an Unknown Parameter in a Distribution**

A discrete random variable X has the following probability distribution:

x	1	2	3	4	5
$P(X = x)$	0.1	0.2	k	0.3	0.1

1. Find the value of k.
2. Write down the full probability distribution.

SOLUTION

Solution.

1. Sum of probabilities must equal 1:

$$0.1 + 0.2 + k + 0.3 + 0.1 = 1 \implies 0.7 + k = 1 \implies k = 0.3.$$

2. The full distribution is:

x	1	2	3	4	5
$P(X = x)$	0.1	0.2	0.3	0.3	0.1

9.3 Expectation, Variance and Standard Deviation

DEFINITION **Expectation (Mean)**

The **expectation** or **mean** of a discrete random variable X is

$$E(X) = \mu = \sum_{x \in S_X} x\, P(X = x),$$

provided the series converges absolutely.

DEFINITION **Variance and Standard Deviation**

The **variance** of X is

$$\operatorname{Var}(X) = \sigma^2 = E\left[(X - \mu)^2\right] = \sum_{x \in S_X} (x - \mu)^2 P(X = x).$$

The **standard deviation** is $\sigma = \sqrt{\operatorname{Var}(X)}$.

PROPERTY **Computational Formula for Variance (Examinable)**

$$\operatorname{Var}(X) = E(X^2) - [E(X)]^2, \quad \text{where } E(X^2) = \sum x^2 P(X = x).$$

PROPERTY **Linear Transformation Properties (Examinable)**

For constants $a, b \in \mathbb{R}$:

$$E(aX + b) = aE(X) + b,$$
$$\text{Var}(aX + b) = a^2\text{Var}(X).$$

In particular, $\text{Var}(X + b) = \text{Var}(X)$ (adding a constant does not change spread).

METHOD **Computing Expectation, Variance and Standard Deviation**

1. Write down the probability distribution of X.
2. Compute $E(X) = \sum x\,P(X = x)$.
3. Compute $E(X^2) = \sum x^2\,P(X = x)$.
4. Compute $\text{Var}(X) = E(X^2) - [E(X)]^2$.
5. Compute $\sigma_X = \sqrt{\text{Var}(X)}$.
6. For linear transformations, use $E(aX + b) = aE(X) + b$ and $\text{Var}(aX + b) = a^2\text{Var}(X)$.

WORKED EXAMPLE 23 **Mean and Variance of a Discrete Random Variable**

A discrete random variable X has the following probability distribution:

x	1	2	3	4	5
$P(X = x)$	0.1	0.2	0.3	0.3	0.1

1. Calculate $E(X)$ and $\text{Var}(X)$.
2. Find $E(2X - 3)$ and $\text{Var}(2X - 3)$.

SOLUTION

Solution.

1.

$$E(X) = \sum xP(X = x) = 1(0.1) + 2(0.2) + 3(0.3) + 4(0.3) + 5(0.1) = 0.1 + 0.4 + 0.9 + 1.2 + 0.5 = 3.1.$$

$$E(X^2) = \sum x^2P(X = x) = 1^2(0.1)+2^2(0.2)+3^2(0.3)+4^2(0.3)+5^2(0.1) = 0.1+0.8+2.7+4.8+2.5 = 10.9.$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 10.9 - (3.1)^2 = 10.9 - 9.61 = 1.29.$$

Standard deviation $\sigma_X = \sqrt{1.29} \approx 1.136$.

2. Using linearity properties:

$$E(2X - 3) = 2E(X) - 3 = 2(3.1) - 3 = 6.2 - 3 = 3.2.$$

$$\text{Var}(2X - 3) = 2^2\text{Var}(X) = 4 \times 1.29 = 5.16.$$

WORKED EXAMPLE 24 **Finding a Parameter and Expectation**

The random variable X takes values 1, 2, 3, 4 with probabilities:

$$P(X = x) = \begin{cases} kx & x = 1, 2 \\ k(5 - x) & x = 3, 4 \\ 0 & \text{otherwise} \end{cases}$$

where k is a constant.

1. Find the value of k.
2. Calculate $E(X)$ and $\text{Var}(X)$.
3. Find $P(X > E(X))$.

SOLUTION

Solution.

1. Sum of probabilities:

$$P(X = 1) = k, \quad P(X = 2) = 2k, \quad P(X = 3) = 2k, \quad P(X = 4) = k.$$

Sum $= k + 2k + 2k + k = 6k = 1 \Rightarrow k = \frac{1}{6}$. Distribution:

x	1	2	3	4
$P(X = x)$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{6}$

2. $E(X) = 1\cdot\frac{1}{6} + 2\cdot\frac{1}{3} + 3\cdot\frac{1}{3} + 4\cdot\frac{1}{6} = \frac{1}{6} + \frac{2}{3} + 1 + \frac{2}{3} = \frac{1}{6} + \frac{4}{6} + \frac{6}{6} + \frac{4}{6} = \frac{15}{6} = 2.5$. $E(X^2) = 1^2\cdot\frac{1}{6} + 4\cdot\frac{1}{3} + 9\cdot\frac{1}{3} + 16\cdot\frac{1}{6} = \frac{1}{6} + \frac{4}{3} + 3 + \frac{8}{3} = \frac{1}{6} + \frac{8}{6} + \frac{18}{6} + \frac{16}{6} = \frac{43}{6}$. $\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{43}{6} - \left(\frac{5}{2}\right)^2 = \frac{43}{6} - \frac{25}{4} = \frac{86}{12} - \frac{75}{12} = \frac{11}{12}$.
3. $E(X) = 2.5$. $P(X > 2.5) = P(X = 3) + P(X = 4) = \frac{1}{3} + \frac{1}{6} = \frac{1}{2}$.

9.4 The Cumulative Distribution Function, Mode and Median

DEFINITION **Cumulative Distribution Function (CDF)**

The **cumulative distribution function** (CDF) of a discrete random variable X is defined for all real x by

$$F(x) = P(X \le x) = \sum_{x_i \le x} P(X = x_i).$$

PROPERTY **Properties of the CDF (Examinable)**

For a discrete RV X:

1. $F(x)$ is a step function, non-decreasing.
2. $F(x)$ is right-continuous: $\lim_{h \to 0^+} F(x + h) = F(x)$.
3. $\lim_{x \to -\infty} F(x) = 0$, $\lim_{x \to +\infty} F(x) = 1$.
4. $P(X = x) = F(x) - F(x^-)$ where $F(x^-) = \lim_{h \to 0^+} F(x - h)$ (the jump at x).
5. For any $a < b$, $P(a < X \le b) = F(b) - F(a)$.

DEFINITION **Mode**

The **mode** of a discrete random variable X is the value x that maximizes $P(X = x)$, i.e., the most probable value. If several values share the maximum probability, the distribution is **multimodal**.

DEFINITION **Median**

A **median** of a discrete random variable X is any value m such that

$$P(X \le m) \ge \frac{1}{2} \quad \text{and} \quad P(X \ge m) \ge \frac{1}{2}.$$

For discrete variables, the median is often taken as the smallest x with $F(x) \ge 0.5$.

METHOD **Cumulative Distribution Function, Median and Mode**

1. For a discrete random variable X, compute $F(x) = P(X \le x)$ for all real x by summing probabilities up to x.

2. $F(x)$ is constant between possible values of X and jumps at each $x_i \in S_X$.
3. Mode: Identify the value(s) with the highest probability $P(X = x)$.
4. Median: Find the smallest x such that $F(x) \geq 0.5$ (or any m satisfying the definition).

WORKED EXAMPLE 25 **CDF, Median and Mode**

The random variable X has probability mass function:

$$P(X = x) = \begin{cases} \frac{x}{10}, & x = 1, 2, 3, 4 \\ 0, & \text{otherwise.} \end{cases}$$

1. Verify that this is a valid probability distribution.
2. Construct the cumulative distribution function $F(x)$.
3. Find the mode and the median of X.

SOLUTION

Solution.

1. $\sum_{x=1}^{4} \frac{x}{10} = \frac{1+2+3+4}{10} = \frac{10}{10} = 1$. All probabilities are non-negative. Valid.
2. Compute $F(x) = P(X \leq x)$:

$$\begin{aligned} F(1) &= P(X \leq 1) = 0.1, \\ F(2) &= P(X \leq 2) = 0.1 + 0.2 = 0.3, \\ F(3) &= P(X \leq 3) = 0.3 + 0.3 = 0.6, \\ F(4) &= P(X \leq 4) = 0.6 + 0.4 = 1. \end{aligned}$$

For $x < 1$, $F(x) = 0$; for $1 \leq x < 2$, $F(x) = 0.1$; for $2 \leq x < 3$, $F(x) = 0.3$; for $3 \leq x < 4$, $F(x) = 0.6$; for $x \geq 4$, $F(x) = 1$. In table form:

x	$(-\infty, 1)$	$[1, 2)$	$[2, 3)$	$[3, 4)$	$[4, \infty)$
$F(x)$	0	0.1	0.3	0.6	1

3. Mode: The highest probability is $P(X = 4) = 0.4$. Mode = 4. Median: Smallest x with $F(x) \geq 0.5$. $F(2) = 0.3 < 0.5$, $F(3) = 0.6 \geq 0.5$. Median = 3.

9.5 The Distribution of $X_1 + X_2$

PROPERTY **Sum of Independent Discrete Random Variables**

Let X and Y be independent discrete random variables. The sum $S = X + Y$ is a discrete random variable with probability mass function given by the **convolution**:

$$P(S = s) = \sum_{x} P(X = x)\,P(Y = s - x),$$

where the sum runs over all x such that $P(X = x) > 0$ and $P(Y = s - x) > 0$.

PROPERTY **Expectation and Variance of a Sum (Examinable)**

For any random variables X, Y (not necessarily independent):

$$E(X + Y) = E(X) + E(Y).$$

If X and Y are independent:

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y).$$

METHOD **Distribution of the Sum of Two Independent Discrete Random Variables**

1. Let X and Y be independent discrete random variables with known distributions.
2. List all pairs (x, y) with their joint probabilities $P(X = x)P(Y = y)$ (by independence).
3. Compute $s = x + y$ for each pair.
4. Group pairs by the value of s and add the probabilities to obtain $P(S = s)$.
5. Alternatively, use the convolution formula directly.
6. Compute $E(S)$ and $\mathrm{Var}(S)$ either from the distribution of S or using the additive properties.

WORKED EXAMPLE 26 **Sum of Two Independent Dice Scores**

Two fair six-sided dice are rolled. Let X be the score on the first die, Y the score on the second. X and Y are independent. Let $S = X + Y$.

1. Find the probability distribution of S.
2. Compute $E(S)$ and $\mathrm{Var}(S)$ directly from the distribution of S, and verify the additive properties.

SOLUTION

Solution.

1. Possible values of S: $2, 3, \ldots, 12$. There are 36 equally likely outcomes (x, y). Count pairs for each sum:

s	2	3	4	5	6	7	8	9	10	11	12
Number of pairs	1	2	3	4	5	6	5	4	3	2	1

Thus $P(S = s) = \frac{\text{number of pairs}}{36}$. Distribution:

s	2	3	4	5	6	7	8	9	10	11	12
$P(S = s)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

2. Compute $E(S)$:

$$E(S) = \sum s\,P(S = s) = \frac{1}{36}\left(2\cdot1 + 3\cdot2 + 4\cdot3 + 5\cdot4 + 6\cdot5 + 7\cdot6 + 8\cdot5 + 9\cdot4 + 10\cdot3 + 11\cdot2 + 12\cdot1\right).$$

Sum $= 2 + 6 + 12 + 20 + 30 + 42 + 40 + 36 + 30 + 22 + 12 = 252$. $E(S) = \frac{252}{36} = 7$. $E(X) = E(Y) = \frac{1+2+3+4+5+6}{6} = 3.5$. $E(X) + E(Y) = 7$. **Verified.**

$E(S^2) = \frac{1}{36}\left(4\cdot1 + 9\cdot2 + 16\cdot3 + 25\cdot4 + 36\cdot5 + 49\cdot6 + 64\cdot5 + 81\cdot4 + 100\cdot3 + 121\cdot2 + 144\cdot1\right)$.
Sum $= 4 + 18 + 48 + 100 + 180 + 294 + 320 + 324 + 300 + 242 + 144 = 1974$. $E(S^2) = \frac{1974}{36} = \frac{329}{6}$.
$\mathrm{Var}(S) = E(S^2) - [E(S)]^2 = \frac{329}{6} - 49 = \frac{329-294}{6} = \frac{35}{6} \approx 5.8333$.
$\mathrm{Var}(X) = \mathrm{Var}(Y) = \frac{35}{12}$ (standard result for a fair die). $\mathrm{Var}(X) + \mathrm{Var}(Y) = \frac{35}{12} + \frac{35}{12} = \frac{35}{6}$. **Verified.**

WORKED EXAMPLE 27 **Sum of Two Independent Non-Identical Distributions**

Let X and Y be independent random variables with distributions:

x	0	1	2
$P(X = x)$	0.2	0.5	0.3

y	1	2
$P(Y = y)$	0.4	0.6

Find the distribution of $S = X + Y$, $E(S)$ and $\mathrm{Var}(S)$.

SOLUTION

Solution. Possible values of S: $1, 2, 3, 4$.

$$P(S=1) = P(X=0, Y=1) = 0.2 \times 0.4 = 0.08,$$
$$P(S=2) = P(X=0, Y=2) + P(X=1, Y=1) = 0.2 \times 0.6 + 0.5 \times 0.4 = 0.12 + 0.20 = 0.32,$$
$$P(S=3) = P(X=1, Y=2) + P(X=2, Y=1) = 0.5 \times 0.6 + 0.3 \times 0.4 = 0.30 + 0.12 = 0.42,$$
$$P(S=4) = P(X=2, Y=2) = 0.3 \times 0.6 = 0.18.$$

Check sum: $0.08 + 0.32 + 0.42 + 0.18 = 1.00$. Distribution of S:

s	1	2	3	4
$P(S=s)$	0.08	0.32	0.42	0.18

$E(X) = 0(0.2) + 1(0.5) + 2(0.3) = 1.1$, $E(Y) = 1(0.4) + 2(0.6) = 1.6$. $E(S) = 1.1 + 1.6 = 2.7$. Check: $1(0.08) + 2(0.32) + 3(0.42) + 4(0.18) = 0.08 + 0.64 + 1.26 + 0.72 = 2.7$. $\text{Var}(X) = E(X^2) - [E(X)]^2 = (0+0.5+1.2) - 1.21 = 1.7 - 1.21 = 0.49$. $\text{Var}(Y) = E(Y^2) - [E(Y)]^2 = (0.4+2.4) - 2.56 = 2.8 - 2.56 = 0.24$. $\text{Var}(S) = 0.49 + 0.24 = 0.73$. Check from S: $E(S^2) = 1(0.08) + 4(0.32) + 9(0.42) + 16(0.18) = 0.08 + 1.28 + 3.78 + 2.88 = 8.02$. $\text{Var}(S) = 8.02 - 2.7^2 = 8.02 - 7.29 = 0.73$.

9.6 Probability Generating Functions

DEFINITION **Probability Generating Function (PGF)**

For a discrete random variable X taking values in $\{0, 1, 2, \ldots\}$ (non-negative integers), the **probability generating function** (PGF) is defined by

$$G_X(t) = E(t^X) = \sum_{k=0}^{\infty} P(X=k)\, t^k,$$

for all real t where the series converges (at least for $|t| \le 1$).

WARNING **Domain of PGF**

The PGF is defined for RVs taking values $0, 1, 2, \ldots$. If X takes other integer values (e.g., negative), we can sometimes shift it ($Y = X + c$) to use PGF, or use the Moment Generating Function (MGF) instead. In GCE Advanced Level, PGF questions typically involve non-negative integer-valued RVs.

PROPERTY **Properties of the PGF (Examinable)**

Let X be a discrete RV with PGF $G_X(t)$.

1. $G_X(1) = \sum P(X = k) = 1$.
2. $G_X(0) = P(X = 0)$.
3. Mean: $E(X) = G_X'(1)$ (first derivative evaluated at $t = 1$).
4. Variance: $\text{Var}(X) = G_X''(1) + G_X'(1) - [G_X'(1)]^2$.
5. Sum of independent variables: If X, Y independent, $G_{X+Y}(t) = G_X(t)G_Y(t)$.
6. Probabilities from PGF: $P(X = k) = \frac{G_X^{(k)}(0)}{k!}$ (coefficient of t^k in the power series expansion).

KEY POINTS **Common PGFs to Memorise**

- Bernoulli(p) ($P(X = 1) = p, P(X = 0) = q = 1 - p$): $G(t) = q + pt$.
- Binomial(n, p): $G(t) = (q + pt)^n$.
- Poisson(λ): $G(t) = e^{\lambda(t-1)}$.
- Geometric(p) on $\{0, 1, 2, \dots\}$ ($P(X = k) = pq^k$): $G(t) = \frac{p}{1-qt}$.

METHOD **Probability Generating Function — Definition and Basic Properties**

1. Write down the PMF $P(X = k)$ for $k = 0, 1, 2, \dots$.
2. Form the series $G_X(t) = \sum_{k=0}^{\infty} P(X = k)t^k$. Simplify to a closed form if possible.
3. Check $G_X(1) = 1$.
4. To find mean: compute $G_X'(t)$, evaluate at $t = 1$.
5. To find variance: compute $G_X''(t)$, evaluate at $t = 1$, use $\text{Var}(X) = G_X''(1) + G_X'(1) - [G_X'(1)]^2$.
6. To find probabilities: expand $G_X(t)$ as a power series; coefficient of t^k is $P(X = k)$.

WORKED EXAMPLE 28 **Using the PGF to Find Mean and Variance**

A discrete random variable X has probability generating function $G_X(t) = \frac{1}{12}(1+t)^2(1+2t)$ for $|t| \le 1$.

1. Find the probability distribution of X.
2. Compute $E(X)$ and $\mathrm{Var}(X)$ using the PGF.

SOLUTION

Solution.

1. Expand $G_X(t)$:

$$(1+t)^2 = 1 + 2t + t^2.$$

$(1+2t+t^2)(1+2t) = 1(1+2t) + 2t(1+2t) + t^2(1+2t) = 1 + 2t + 2t + 4t^2 + t^2 + 2t^3 = 1 + 4t + 5t^2 + 2t^3.$

Multiply by $\frac{1}{12}$:

$$G_X(t) = \frac{1}{12} + \frac{4}{12}t + \frac{5}{12}t^2 + \frac{2}{12}t^3 = \frac{1}{12} + \frac{1}{3}t + \frac{5}{12}t^2 + \frac{1}{6}t^3.$$

Since $G_X(t) = \sum P(X = k)t^k$, we read off:

$$P(X=0) = \frac{1}{12}, \quad P(X=1) = \frac{1}{3}, \quad P(X=2) = \frac{5}{12}, \quad P(X=3) = \frac{1}{6}.$$

Check sum: $\frac{1}{12} + \frac{4}{12} + \frac{5}{12} + \frac{2}{12} = \frac{12}{12} = 1$. Valid.

2. Compute derivatives:

$$G_X'(t) = \frac{1}{3} + \frac{10}{12}t + \frac{3}{6}t^2 = \frac{1}{3} + \frac{5}{6}t + \frac{1}{2}t^2.$$

$$G_X''(t) = \frac{5}{6} + t.$$

Evaluate at $t = 1$:

$$G_X'(1) = \frac{1}{3} + \frac{5}{6} + \frac{1}{2} = \frac{2}{6} + \frac{5}{6} + \frac{3}{6} = \frac{10}{6} = \frac{5}{3}.$$

$$G_X''(1) = \frac{5}{6} + 1 = \frac{11}{6}.$$

Then $E(X) = G_X'(1) = \frac{5}{3}$. $\mathrm{Var}(X) = G_X''(1) + G_X'(1) - [G_X'(1)]^2 = \frac{11}{6} + \frac{5}{3} - \left(\frac{5}{3}\right)^2 = \frac{11}{6} + \frac{10}{6} - \frac{25}{9} = \frac{21}{6} - \frac{25}{9} = \frac{7}{2} - \frac{25}{9} = \frac{63}{18} - \frac{50}{18} = \frac{13}{18}$.

METHOD **Probability Generating Function of the Sum of Independent Random Variables**

1. If $X_1, X_2, \ldots, X_n$ are independent, then $G_{X_1+\cdots+X_n}(t) = G_{X_1}(t)G_{X_2}(t)\cdots G_{X_n}(t)$.

2. To find the distribution of the sum:
   - Expand the product PGF and read coefficients (if simple).
   - Use the convolution method (Method 5).
   - Recognise the resulting PGF as that of a known distribution (e.g., sum of independent Poissons is Poisson with parameter sum of parameters; sum of independent Binomials with same p is Binomial).
3. Mean and variance of the sum are sums of individual means and variances.

WORKED EXAMPLE 29 **Sum of Independent Poisson Variables via PGF**

The number of customers arriving at a shop in Yaoundé during the morning follows a Poisson distribution with mean 3, and the number arriving in the afternoon follows an independent Poisson distribution with mean 5. Let X be the morning arrivals, Y the afternoon arrivals.

1. Write down the PGFs of X and Y.
2. Find the PGF of $T = X + Y$ (total daily arrivals).
3. Identify the distribution of T and state its mean and variance.
4. Find $P(T = 7)$.

SOLUTION

Solution.

1. For Poisson(λ), $G(t) = e^{\lambda(t-1)}$. $G_X(t) = e^{3(t-1)}$, $G_Y(t) = e^{5(t-1)}$.
2. $G_T(t) = G_X(t)G_Y(t) = e^{3(t-1)}e^{5(t-1)} = e^{(3+5)(t-1)} = e^{8(t-1)}$.
3. $G_T(t)$ is the PGF of a Poisson distribution with parameter $\lambda = 8$. Thus $T \sim$ Poisson(8). Mean $E(T) = 8$, Variance $\mathrm{Var}(T) = 8$.
4. $P(T = 7) = \frac{e^{-8}8^7}{7!}$. Compute numerically: $8^7 = 2097152$, $7! = 5040$, $\frac{2097152}{5040} \approx 416.1$, $e^{-8} \approx 0.00033546$, product ≈ 0.1396.

WORKED EXAMPLE 30 **Sum of Two Independent Dice Scores via PGF**

Two fair six-sided dice are rolled. Let X and Y be the scores. Find the PGF of X, the PGF of $S = X + Y$, and hence the probability $P(S = 7)$.

SOLUTION

Solution. X takes values $1, \ldots, 6$ with probability $\frac{1}{6}$ each. But PGF requires values $0, 1, 2, \ldots$. We can define $X' = X - 1$ (values $0, \ldots, 5$) or use the PGF definition $G_X(t) = E(t^X) = \frac{1}{6}(t + t^2 + t^3 + t^4 + t^5 + t^6)$ which is valid for any integer-valued RV as a formal power series. The properties $G_{X+Y} = G_X G_Y$ and $E(X) = G_X'(1)$ still hold for integer-valued RVs (with $G_X(t) = \sum P(X = k)t^k$).

$$G_X(t) = \frac{t(1 - t^6)}{6(1 - t)} = \frac{1}{6}(t + t^2 + t^3 + t^4 + t^5 + t^6).$$

Since X, Y independent and identically distributed:

$$G_S(t) = [G_X(t)]^2 = \frac{1}{36}(t + t^2 + t^3 + t^4 + t^5 + t^6)^2.$$

We need coefficient of t^7 in the expansion. The pairs (i, j) with $i + j = 7$, $1 \le i, j \le 6$ are $(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)$ — 6 pairs. Each pair contributes $\frac{1}{36}t^7$. Coefficient $= 6 \times \frac{1}{36} = \frac{1}{6}$. Thus $P(S = 7) = \frac{1}{6}$.

9.7 Comprehensive Applications

METHOD **Comprehensive Problem: Game of Chance with Discrete Random Variable**

1. Model the game: define the random variable (net gain/loss or winnings).
2. Construct its probability distribution (account for all outcomes).
3. Compute expectation (fair game if $E = 0$).
4. Compute variance/standard deviation for risk assessment.
5. If sums of independent plays are considered, use additive properties of expectation/variance or PGF (if the RV is non-negative integer valued).

WORKED EXAMPLE 31 **A Game at a Cameroon Fair**

At a fair in Douala, a game consists of drawing a ball from a bag containing 4 red, 3 blue, and 3 green balls. You pay 500 FCFA to play. If you draw a red ball, you win 2000 FCFA; if blue, you win 1000 FCFA; if green, you win nothing. Let W be your net gain (winnings minus cost).

1. Find the probability distribution of W.
2. Calculate $E(W)$ and $\mathrm{Var}(W)$. Is the game fair?

3. If you play twice independently, let $T = W_1 + W_2$ be your total net gain. Find $E(T)$ and $\text{Var}(T)$.

4. Find the probability that your total net gain after two games is exactly 1000 FCFA.

SOLUTION

Solution.

1. Probabilities: $P(\text{Red}) = \frac{4}{10} = 0.4$, $P(\text{Blue}) = \frac{3}{10} = 0.3$, $P(\text{Green}) = \frac{3}{10} = 0.3$. Net gains: Red: 2000 − 500 = 1500 FCFA. Blue: 1000 − 500 = 500 FCFA. Green: 0 − 500 = −500 FCFA. Distribution of W:

w	1500	500	−500
$P(W = w)$	0.4	0.3	0.3

2. $E(W) = 1500(0.4) + 500(0.3) + (-500)(0.3) = 600 + 150 - 150 = 600$ FCFA. $E(W^2) = 1500^2(0.4) + 500^2(0.3) + (-500)^2(0.3) = 2250000(0.4) + 250000(0.3) + 250000(0.3) = 900000 + 75000 + 75000 = 1050000$. $\text{Var}(W) = 1050000 - 600^2 = 1050000 - 360000 = 690000$ FCFA^2. $\sigma_W = \sqrt{690000} \approx 830.66$ FCFA. Since $E(W) = 600 > 0$, the game is not fair; it favours the player (on average you gain 600 FCFA per game).

3. For two independent games, $T = W_1 + W_2$. $E(T) = E(W_1) + E(W_2) = 600 + 600 = 1200$ FCFA. $\text{Var}(T) = \text{Var}(W_1) + \text{Var}(W_2) = 690000 + 690000 = 1380000$ FCFA^2.

4. We need $P(T = 1000)$. Possible pairs of outcomes:

   - (Red, Green) or (Green, Red): net gain $1500 + (-500) = 1000$. Probability $2 \times 0.4 \times 0.3 = 0.24$.
   - (Blue, Blue): net gain $500 + 500 = 1000$. Probability $0.3 \times 0.3 = 0.09$.

   No other combinations give 1000. Total $P(T = 1000) = 0.24 + 0.09 = 0.33$.

DID YOU KNOW? **PGF and the Douala Fair Game**

The net gain W takes negative values, so its standard PGF $G_W(t) = E(t^W)$ would involve negative powers of t (a Laurent series). While the convolution property $G_{W_1+W_2} = G_{W_1} G_{W_2}$ still holds formally, the standard PGF theory (derivatives at $t = 1$ giving moments) applies directly only to non-negative integer-valued RVs. For W, we used the direct convolution method (listing pairs) which is always valid. In exams, if a PGF question involves a RV with negative values, shift it to make it non-negative (e.g., define $U = W + 500$ taking values 0, 1000, 2000) or use the MGF $M_X(t) = E(e^{tX})$.

9.8 Exercises

EXERCISE 8 **Exercise 1**

A discrete random variable X has probability function $P(X = x) = \frac{c}{x(x+1)}$ for $x = 1, 2, 3, 4, 5$.

1. Find the value of the constant c.
2. Calculate $E(X)$ and $\mathrm{Var}(X)$.
3. Find the cumulative distribution function $F(x)$ and sketch its graph.
4. Determine the mode and median of X.

EXERCISE 9 **Exercise 2**

Two independent random variables X and Y have the following distributions:

x	0	1	2
$P(X = x)$	0.3	0.4	0.3

y	0	1
$P(Y = y)$	0.6	0.4

1. Find the probability distribution of $S = X + Y$.
2. Compute $E(S)$ and $\mathrm{Var}(S)$ using the additive properties.
3. Verify your answers by computing directly from the distribution of S.

EXERCISE 10 **Exercise 3**

A random variable X has PGF $G_X(t) = \frac{1}{8}(1 + t)^3$.

1. Identify the distribution of X and its parameters.
2. Find $E(X)$ and $\mathrm{Var}(X)$ using the PGF.
3. Find $P(X \geq 2)$.

EXERCISE 11 **Exercise 4**

The number of emails received per hour by a server follows a Poisson distribution with mean 4. The number of spam emails per hour follows an independent Poisson distribution with mean 1. Let T be the total number of emails received in a 2-hour period.

1. Find the distribution of T.

2. Calculate $P(T = 10)$.
3. Find the probability that exactly 3 of the 10 emails are spam, given $T = 10$. (Hint: conditional distribution of Poisson components).

EXERCISE 12 **Exercise 5**

A biased tetrahedral die (faces 1,2,3,4) has probabilities $P(1) = 0.1$, $P(2) = 0.2$, $P(3) = 0.3$, $P(4) = 0.4$. The die is rolled twice. Let M be the maximum of the two scores.

1. Find the probability distribution of M.
2. Calculate $E(M)$ and $\text{Var}(M)$.
3. Find the probability generating function of M (treating values as $1, 2, 3, 4$).

10 EXERCISES

10.1 I check my knowledge

EXERCISE 13 **MCQ: spotting a valid distribution**

A discrete random variable X takes the values $0, 1, 2, 3$. In each case, state whether the table can define a probability distribution of X. Justify your answer.

1. $P(X = x) = \dfrac{x+1}{10}$ for $x = 0, 1, 2, 3$.
2. $P(X = 0) = 0.2$, $P(X = 1) = 0.3$, $P(X = 2) = 0.6$, $P(X = 3) = -0.1$.
3. $P(X = x) = \dfrac{1}{2^{x+1}}$ for $x = 0, 1, 2, 3$.
4. $P(X = x) = 0.25$ for $x = 0, 1, 2, 3$.

EXERCISE 14 **True or false?**

Answer true or false to each statement, justifying your answer with a definition, a property or a counter-example.

1. If X is a discrete random variable, then $\sum_x P(X = x) = 1$.
2. For any random variable X, $\mathrm{Var}(X) = E(X^2) - E(X)^2$.
3. The median of a discrete random variable is always one of the values taken by X.
4. If $G_X(t)$ is the probability generating function of X, then $G_X(1) = E(X)$.
5. If X_1 and X_2 are independent, then $E(X_1 + X_2) = E(X_1) + E(X_2)$.

EXERCISE 15 **Definitions in your own words**

1. Define a **random variable** and explain the difference between a discrete and a continuous random variable, giving one example of each.
2. Define the **cumulative distribution function** $F(x)$ of a discrete random variable X and state two of its properties.
3. Define the **probability generating function** of a discrete random variable X taking values in $\mathbb{N}$, and state how $E(X)$ is obtained from it.

EXERCISE 16 **Direct calculations**

A discrete random variable X has distribution

x	1	2	3	4
$P(X = x)$	0.1	0.3	0.4	0.2

1. Verify that this is a valid probability distribution.
2. Calculate $E(X)$, $E(X^2)$ and $\mathrm{Var}(X)$.
3. Write down $F(2)$ and $F(3.5)$.
4. Determine the mode of X.

10.2 I practise

EXERCISE 17 **Finding the constant k**

A discrete random variable X takes the values $1, 2, 3, 4$ with

$$P(X = x) = kx, \qquad x = 1, 2, 3, 4.$$

1. Find the value of the constant k.
2. Tabulate the probability distribution of X.
3. Calculate $E(X)$ and $\mathrm{Var}(X)$.
4. Calculate $P(X \geq 3)$ and $P(X > 1 \mid X \leq 3)$.

EXERCISE 18 **Recovering the mass function from F**

The cumulative distribution function of a discrete random variable X is given by

x	1	2	3	4	5	6
$F(x)$	0.05	0.20	0.45	0.70	0.90	1

1. Recover the probability mass function of X and present it in a table.
2. Find the median of X.
3. Calculate $P(2 < X \leq 5)$.
4. Calculate $E(X)$.

EXERCISE 19 **A game in FCFA**

A game costs 500 FCFA to play. A player draws one ball at random from a bag containing 3 red balls, 2 blue balls and 1 green ball. The player wins 1000 FCFA for a red ball, 2000 FCFA for a blue ball and 5000 FCFA for a green ball. Let G be the player's net gain (winnings minus the entry fee).

1. Construct the probability distribution of G.
2. Compute the expected gain $E(G)$. Is the game fair? Is it favourable to the player or to the organiser?
3. Find the entry fee that would make the game fair.

EXERCISE 20 **Using a p.g.f.**

The probability generating function of a discrete random variable X is

$$G(t) = \frac{1}{8}(1 + t)^3.$$

1. Expand $G(t)$ and hence write down the probability distribution of X.
2. Find $P(X = 2)$.
3. Calculate $G'(1)$ and $G''(1)$, and deduce $E(X)$ and $\mathrm{Var}(X)$.
4. Describe a simple experiment (involving coins) whose outcome has this distribution.

10.3 I prepare for the GCE

EXERCISE 21 **Distribution with an unknown probability** **(12 marks)**

The probability distribution of a discrete random variable X is given by

x	0	1	2	3	4
$P(X = x)$	0.1	p	0.3	0.2	q

where p and q are constants, and it is known that $E(X) = 2.1$.

1. Write down two equations satisfied by p and q, and hence find their values. (4 marks)
2. Calculate $\mathrm{Var}(X)$ and the standard deviation of X, giving your answers to 3 significant figures. (4 marks)
3. Construct the cumulative distribution function $F(x)$ in a table, and state the median and the mode of X. (2 marks)

4. Two independent observations of X are made. Find the probability that their sum equals 4. (2 marks)

EXERCISE 22 **Sum of two dice** **(13 marks)**

A fair tetrahedral die has faces numbered 1, 2, 3, 4; when it is thrown, the score X_1 is the number on the face on which it lands, each face being equally likely. A biased ordinary die has score X_2 with distribution

x	1	2	3	4	5	6
$P(X_2 = x)$	0.1	0.1	0.2	0.2	0.2	0.2

The two dice are thrown independently and $S = X_1 + X_2$ is the total score.

1. Show that $E(X_1) = 2.5$ and find $E(X_2)$. (3 marks)
2. Construct the probability distribution of S in a table. (4 marks)
3. Find the mode and the median of S. (2 marks)
4. Verify that $E(S) = E(X_1) + E(X_2)$. (2 marks)
5. Find $P(S \geq 8)$. (2 marks)

EXERCISE 23 **Comparing two investments** **(15 marks)**

A trader in Douala can invest the same capital in one of two projects, A or B. The annual profit, in hundreds of thousands of FCFA, is modelled by the random variables X_A and X_B with distributions

x	-2	1	3	6
$P(X_A = x)$	0.1	0.3	0.4	0.2
$P(X_B = x)$	0.05	0.25	0.45	0.25

1. Calculate $E(X_A)$ and $E(X_B)$. (4 marks)
2. Calculate the standard deviation of each of X_A and X_B, giving your answers to 3 significant figures. (5 marks)
3. The probability generating function of X_A is $G_A(t)$. Write down $G_A(t)$ as an expression in t and verify that $G_A'(1) = E(X_A)$. (3 marks)
4. A risk-averse client wants a good expected profit but dislikes uncertainty. Using your answers to parts (a) and (b), advise the client which project to choose, justifying your

recommendation clearly. (3 marks)

11 SOLUTIONS TO THE EXERCISES

SOLUTION **Exercise 1 — MCQ: spotting a valid distribution**

Recall that a table defines a probability distribution of a discrete random variable X if and only if (i) every probability satisfies $0 \le P(X = x) \le 1$, and (ii) $\sum_x P(X = x) = 1$. Both conditions must be checked.

1. The probabilities are $\frac{1}{10}, \frac{2}{10}, \frac{3}{10}, \frac{4}{10}$, all between 0 and 1, and
$$\frac{1+2+3+4}{10} = \frac{10}{10} = 1.$$
Valid distribution.
2. $P(X = 3) = -0.1 < 0$: a probability can never be negative. (Note that the sum is 1.0, so condition (ii) holds, but condition (i) fails — one single bad value is enough to reject the table.) Not valid.
3. The probabilities are $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}$, all in $[0, 1]$, but
$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} = \frac{8+4+2+1}{16} = \frac{15}{16} \neq 1.$$
Not valid (condition (ii) fails: the total probability is not 1).
4. Each probability is $0.25 \in [0, 1]$ and $4 \times 0.25 = 1$. Valid distribution (this is the discrete uniform distribution on $\{0, 1, 2, 3\}$).

SOLUTION **Exercise 2 — True or false?**

1. True. The events $\{X = x\}$, as x runs through the values taken by X, form a partition of the sample space Ω (they are mutually exclusive and their union is Ω). Hence the total probability is $\sum_x P(X = x) = P(\Omega) = 1$. This is the fundamental condition satisfied by every probability distribution.
2. True. This is the computational (König–Huygens) formula for the variance. Using the linearity of expectation with $\mu = E(X)$:
$$\begin{aligned}\mathrm{Var}(X) &= E\left[(X - \mu)^2\right] = E(X^2 - 2\mu X + \mu^2) \\ &= E(X^2) - 2\mu\, E(X) + \mu^2 = E(X^2) - 2\mu^2 + \mu^2 = E(X^2) - [E(X)]^2.\end{aligned}$$
3. False. Counter-example: let X take the values 1 and 2, each with probability 0.5. Then $F(1) = 0.5$ and $F(x) = 1$ for all $x \ge 2$. With the midpoint convention, every $m \in [1, 2]$

satisfies $P(X \le m) \ge 0.5$ and $P(X \ge m) \ge 0.5$, and one usually takes the median to be $m = 1.5$, which is not a value taken by X. (With the GCE convention "the median is the smallest value x such that $F(x) \ge 0.5$", the median here is 1, which is a value of X; the statement is false in general because under the midpoint convention the median need not belong to the set of values of X.)

4. False. By definition $G_X(t) = E(t^X) = \sum_x t^x P(X = x)$, so substituting $t = 1$ gives $G_X(1) = \sum_x P(X = x) = 1$, not $E(X)$. It is the derivative evaluated at $t = 1$ that gives the mean: $G'_X(1) = E(X)$.

5. True. Expectation is additive for any two random variables, independent or not: $E(X_1 + X_2) = E(X_1) + E(X_2)$. Independence is required for the variance formula $\mathrm{Var}(X_1 + X_2) = \mathrm{Var}(X_1) + \mathrm{Var}(X_2)$; when independence holds, the expectation formula holds a fortiori.

SOLUTION **Exercise 3 — Definitions in your own words**

1. A **random variable** is a real-valued function defined on the sample space Ω of a random experiment: to each outcome $\omega \in \Omega$ it associates a real number $X(\omega)$. A random variable is **discrete** if it takes only finitely many or countably many values (e.g. the number of goals scored by a football team in a match, taking values in $\mathbb{N}$). It is **continuous** if it can take any value in an interval of $\mathbb{R}$ (e.g. the height of a randomly chosen student in a lycée in Yaoundé, taking values in an interval such as $[1.40, 2.00]$ metres).

2. The **cumulative distribution function** of X is the function F defined for every real number x by $F(x) = P(X \le x)$. Two properties: (i) F is non-decreasing (if $a \le b$ then $F(a) \le F(b)$, because the event $\{X \le a\}$ is contained in $\{X \le b\}$); (ii) $F(x) \to 0$ as $x \to -\infty$ and $F(x) \to 1$ as $x \to +\infty$. For a discrete variable, F is a step function: it is constant between two consecutive values of X, it is right-continuous, and it has a jump of height $P(X = x)$ at each value x taken by X.

3. For a discrete random variable X taking values in $\mathbb{N}$, the **probability generating function** is
$$G_X(t) = E(t^X) = \sum_{x=0}^{+\infty} P(X = x)\, t^x.$$
Differentiating term by term, $G'_X(t) = \sum_x x\, P(X = x)\, t^{x-1}$, so setting $t = 1$ gives $G'_X(1) = \sum_x x\, P(X = x)$, that is
$$E(X) = G'_X(1).$$
(One also obtains the variance from $\mathrm{Var}(X) = G''_X(1) + G'_X(1) - [G'_X(1)]^2$.)

SOLUTION **Exercise 4 — Direct calculations**

1. All four probabilities lie in $[0, 1]$ and

$$0.1 + 0.3 + 0.4 + 0.2 = 1.0.$$

Both conditions are satisfied, hence the table defines a valid probability distribution.

2.

$$E(X) = 1(0.1) + 2(0.3) + 3(0.4) + 4(0.2) = 0.1 + 0.6 + 1.2 + 0.8 = 2.7.$$
$$E(X^2) = 1^2(0.1) + 2^2(0.3) + 3^2(0.4) + 4^2(0.2) = 0.1 + 1.2 + 3.6 + 3.2 = 8.1.$$
$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = 8.1 - (2.7)^2 = 8.1 - 7.29 = 0.81.$$

The standard deviation is $\sigma = \sqrt{\mathrm{Var}(X)} = \sqrt{0.81} = 0.9$.

3. $F(2) = P(X \le 2) = P(X = 1) + P(X = 2) = 0.1 + 0.3 = 0.4$. Since X takes no value strictly between 3 and 4, the event $\{X \le 3.5\}$ is the same as $\{X \le 3\}$, so

$$F(3.5) = P(X \le 3.5) = P(X \le 3) = 0.1 + 0.3 + 0.4 = 0.8.$$

4. The mode is the value with the largest probability. The largest probability is 0.4, attained at $x = 3$. The mode of X is 3.

SOLUTION **Exercise 5 — Finding the constant k**

1. The total probability must equal 1:

$$\sum_{x=1}^{4} kx = k(1 + 2 + 3 + 4) = 10k = 1 \implies k = \frac{1}{10} = 0.1.$$

(Note that $k = 0.1 \ge 0$, so all the probabilities kx are non-negative, as required.)

2. The distribution of X:

x	1	2	3	4
$P(X = x)$	0.1	0.2	0.3	0.4

3.

$$E(X) = 1(0.1) + 2(0.2) + 3(0.3) + 4(0.4) = 0.1 + 0.4 + 0.9 + 1.6 = 3.0.$$
$$E(X^2) = 1^2(0.1) + 2^2(0.2) + 3^2(0.3) + 4^2(0.4) = 0.1 + 0.8 + 2.7 + 6.4 = 10.0.$$
$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = 10.0 - (3.0)^2 = 10 - 9 = 1.0.$$

4. $P(X \ge 3) = P(X = 3) + P(X = 4) = 0.3 + 0.4 = 0.7$.

For the conditional probability, using $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$ with $A = \{X > 1\}$ and $B = \{X \le 3\}$, so that $A \cap B = \{1 < X \le 3\} = \{X = 2\} \cup \{X = 3\}$:

$$P(X > 1 \mid X \le 3) = \frac{P(1 < X \le 3)}{P(X \le 3)} = \frac{P(X = 2) + P(X = 3)}{P(X = 1) + P(X = 2) + P(X = 3)} = \frac{0.2 + 0.3}{0.1 + 0.2 + 0.3} = \frac{0.5}{0.6} = \frac{5}{6}.$$

SOLUTION **Exercise 6 — Recovering the mass function from F**

1. For a discrete variable taking integer values, each jump of F gives a probability: $P(X = x) = F(x) - F(x - 1)$. Here the smallest value is 1, so $F(0) = 0$:

$$P(X = 1) = F(1) - F(0) = 0.05 - 0 = 0.05,$$
$$P(X = 2) = F(2) - F(1) = 0.20 - 0.05 = 0.15,$$
$$P(X = 3) = F(3) - F(2) = 0.45 - 0.20 = 0.25,$$
$$P(X = 4) = F(4) - F(3) = 0.70 - 0.45 = 0.25,$$
$$P(X = 5) = F(5) - F(4) = 0.90 - 0.70 = 0.20,$$
$$P(X = 6) = F(6) - F(5) = 1 - 0.90 = 0.10.$$

x	1	2	3	4	5	6
$P(X = x)$	0.05	0.15	0.25	0.25	0.20	0.10

Check: $0.05 + 0.15 + 0.25 + 0.25 + 0.20 + 0.10 = 1$. ✓

2. The median is the smallest value x such that $F(x) \ge 0.5$. We have $F(3) = 0.45 < 0.5$ and $F(4) = 0.70 \ge 0.5$, so the median is 4.

3. Using $P(a < X \le b) = F(b) - F(a)$:

$$P(2 < X \le 5) = F(5) - F(2) = 0.90 - 0.20 = 0.70.$$

(Check directly: $P(X = 3) + P(X = 4) + P(X = 5) = 0.25 + 0.25 + 0.20 = 0.70$. ✓)

4.

$$E(X) = 1(0.05) + 2(0.15) + 3(0.25) + 4(0.25) + 5(0.20) + 6(0.10)$$
$$= 0.05 + 0.30 + 0.75 + 1.00 + 1.00 + 0.60 = 3.70.$$

SOLUTION **Exercise 7 — A game in FCFA**

1. There are $3 + 2 + 1 = 6$ equiprobable balls. The net gain is G = winnings − 500:
   - Red ball: $G = 1000 - 500 = 500$ FCFA, with probability $\frac{3}{6} = \frac{1}{2}$;
   - Blue ball: $G = 2000 - 500 = 1500$ FCFA, with probability $\frac{2}{6} = \frac{1}{3}$;
   - Green ball: $G = 5000 - 500 = 4500$ FCFA, with probability $\frac{1}{6}$.

g (FCFA)	500	1500	4500
$P(G = g)$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{6}$

   (Check: $\frac{1}{2} + \frac{1}{3} + \frac{1}{6} = \frac{3+2+1}{6} = 1$. ✓)

2. $$E(G) = 500 \times \frac{1}{2} + 1500 \times \frac{1}{3} + 4500 \times \frac{1}{6} = 250 + 500 + 750 = 1500 \text{ FCFA}.$$
   A game is fair when the expected net gain is zero. Since $E(G) = +1500 \neq 0$, the game is not fair. As $E(G) > 0$, on average the player wins 1500 FCFA per game: the game is favourable to the player (and unfavourable to the organiser — such a generous organiser would soon go bankrupt!).

3. Let c be the new entry fee in FCFA. The net gain becomes $G' = G + 500 - c$ (the player now pays c instead of 500), so by linearity of expectation,
   $$E(G') = E(G) + 500 - c = 1500 + 500 - c = 2000 - c.$$
   The game is fair when $E(G') = 0$, i.e. $c = 2000$ FCFA.
   Equivalently, the fair price equals the expected winnings W: using exact fractions,
   $$E(W) = 1000\times\frac{1}{2}+2000\times\frac{1}{3}+5000\times\frac{1}{6} = 500+\frac{2000}{3}+\frac{2500}{3} = 500+\frac{4500}{3} = 500+1500 = 2000 \text{ FCFA}.$$

SOLUTION **Exercise 8 — Using a p.g.f.**

1. Expand with the binomial theorem: $(1 + t)^3 = 1 + 3t + 3t^2 + t^3$, so
   $$G(t) = \frac{1}{8}(1 + t)^3 = \frac{1}{8} + \frac{3}{8}t + \frac{3}{8}t^2 + \frac{1}{8}t^3.$$
   Since $G(t) = \sum_x P(X = x)\, t^x$, the coefficient of t^x is exactly $P(X = x)$. Reading off the coefficients:

x	0	1	2	3
$P(X = x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Check: $\frac{1+3+3+1}{8} = \frac{8}{8} = 1$. ✓

2. $P(X = 2)$ is the coefficient of t^2: $P(X = 2) = \frac{3}{8} = 0.375$.

3. Differentiate twice:

$$G'(t) = \frac{3}{8}(1+t)^2, \qquad G''(t) = \frac{6}{8}(1+t) = \frac{3}{4}(1+t).$$

Evaluating at $t = 1$:

$$G'(1) = \frac{3}{8} \times (1+1)^2 = \frac{3}{8} \times 4 = \frac{3}{2} = 1.5, \qquad G''(1) = \frac{3}{4} \times 2 = \frac{3}{2} = 1.5.$$

Therefore

$$E(X) = G'(1) = 1.5,$$

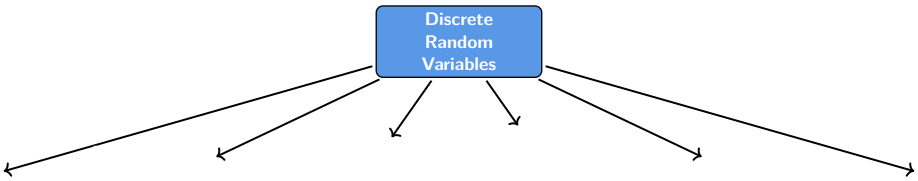
$$\mathrm{Var}(X) = G''(1) + G'(1) - [G'(1)]^2 = 1.5 + 1.5 - (1.5)^2 = 3 - 2.25 = 0.75.$$

(Check by direct computation: $E(X) = 0 \times \frac{1}{8} + 1 \times \frac{3}{8} + 2 \times \frac{3}{8} + 3 \times \frac{1}{8} = \frac{3+6+3}{8} = \frac{12}{8} = 1.5$; $E(X^2) = 0 + \frac{3}{8} + 4 \times \frac{3}{8} + 9 \times \frac{1}{8} = \frac{3+12+9}{8} = 3$, so $\mathrm{Var}(X) = 3 - 1.5^2 = 0.75$. ✓)

4. Toss three fair coins and let X be the number of heads obtained. There are $2^3 = 8$ equiprobable outcomes, and the number of outcomes giving exactly x heads is $\binom{3}{x}$, so $P(X = x) = \binom{3}{x}/8$, i.e. $\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8}$ for $x = 0, 1, 2, 3$ — exactly the distribution found above. (This is the binomial distribution $B(3, \frac{1}{2})$, whose p.g.f. is $(\frac{1}{2} + \frac{1}{2}t)^3 = \frac{1}{8}(1+t)^3$.)

12 SUMMARY SHEET

12.1 Summary Sheet



Chapter PMS11 Mind Map: Discrete Random Variables

KEY POINTS **Key Definitions**

- **Random Variable (RV)**: A function $X : \Omega \to \mathbb{R}$ mapping each outcome of a random experiment to a real number. We denote the RV by a capital letter (X) and its values by small letters (x).
- **Discrete RV**: Takes a finite or countably infinite set of values $\{x_1, x_2, \dots\}$. The set of all possible values is the **support** R_X.
- **Probability Mass Function (PMF)**: $p(x) = P(X = x)$ for $x \in R_X$. Must satisfy:
  1. $p(x) \geq 0$ for all $x \in R_X$,
  2. $\sum_{x \in R_X} p(x) = 1$.
- **Cumulative Distribution Function (CDF)**: $F(x) = P(X \leq x) = \sum_{x_i \leq x} p(x_i)$, defined for all $x \in \mathbb{R}$. Properties: non-decreasing, right-continuous, $F(-\infty) = 0$, $F(+\infty) = 1$.
- **Mode**: The value(s) $x \in R_X$ where $p(x)$ attains its maximum. A distribution may have one mode (unimodal) or several.
- **Median**: Any value m such that $P(X \leq m) \geq 0.5$ and $P(X \geq m) \geq 0.5$. For discrete RVs, the conventional median is the smallest x with $F(x) \geq 0.5$.
- **Probability Generating Function (PGF)**: $G_X(t) = E(t^X) = \sum_{x \in R_X} p(x)t^x$, valid for $|t| \leq 1$. The PGF uniquely determines the distribution.

PROPERTY Essential Laws, Formulas & Theorems

Examinable proofs are marked with an asterisk (*).

- Normalisation: $\sum_{x \in R_X} p(x) = 1$. *Proof: follows from the axioms of probability (total probability of the sample space is 1).*
- Expectation: $E(X) = \sum_{x \in R_X} x\,p(x)$. For any function g, $E(g(X)) = \sum g(x)\,p(x)$ (Law of the Unconscious Statistician).
- Linearity of Expectation: $E(aX + b) = aE(X) + b$ for constants a, b. $E(X + Y) = E(X) + E(Y)$ always holds, independence not required.
- Variance: $Var(X) = E[(X - \mu)^2] = E(X^2) - [E(X)]^2$, where $\mu = E(X)$. Standard deviation $\sigma_X = \sqrt{Var(X)}$.
- Variance Properties: $Var(aX + b) = a^2 Var(X)$. *If X and Y are independent:* $Var(X + Y) = Var(X) + Var(Y)$.
- CDF Properties:

$$P(a < X \le b) = F(b) - F(a),$$
$$P(X = x) = F(x) - F(x^-) \quad \text{(jump at } x\text{)}.$$

- Sum of Independent RVs (Convolution): If X_1, X_2 independent,

$$P(X_1 + X_2 = z) = \sum_{x \in R_{X_1}} P(X_1 = x)\,P(X_2 = z - x).$$

- PGF Properties:

$$G_X(1) = 1 \quad \text{(check)},$$
$$G'_X(1) = E(X),$$
$$G''_X(1) = E[X(X - 1)] = E(X^2) - E(X),$$
$$Var(X) = G''_X(1) + G'_X(1) - [G'_X(1)]^2.$$

- PGF of Sum: If X, Y independent, $G_{X+Y}(t) = G_X(t)G_Y(t)$. *Proof: $E(t^{X+Y}) = E(t^X t^Y) = E(t^X)E(t^Y)$ by independence.*
- Uniqueness: The PGF uniquely determines the probability distribution (examinable concept).

METHOD Standard Methods (Step-by-Step)

1. Constructing a Distribution Table:

(a) Define the RV clearly: "Let X be the number of..."

(b) List all possible values $x \in R_X$.

(c) Compute $p(x) = P(X = x)$ using combinatorics, given probabilities, or a formula.

(d) Verify $\sum p(x) = 1$ explicitly (write the sum and conclude = 1).

(e) Add columns for $xp(x)$ and $x^2p(x)$; compute totals for $E(X)$ and $E(X^2)$.

2. Finding Unknown Constants (e.g., k):

(a) Write the PMF with the unknown parameter(s).

(b) Use $\sum p(x) = 1$ to form an equation.

(c) Solve for the parameter. Check that all $p(x) \geq 0$.

3. Calculating $E(X)$ and $Var(X)$:

(a) Extend the table with columns $xp(x)$ and $x^2p(x)$.

(b) $\mu = \sum xp(x)$.

(c) $E(X^2) = \sum x^2p(x)$.

(d) $Var(X) = E(X^2) - \mu^2$; $\sigma = \sqrt{Var(X)}$.

4. Linear Transform $Y = aX + b$:

(a) Support $R_Y = \{ax + b : x \in R_X\}$.

(b) $p_Y(y) = p_X((y - b)/a)$.

(c) $E(Y) = a\mu + b$, $Var(Y) = a^2\sigma^2$.

5. CDF, Mode, Median:

(a) Compute cumulative sums for $F(x)$ in a table.

(b) Mode = x with highest $p(x)$.

(c) Median = smallest x such that $F(x) \geq 0.5$. (If $F(x) = 0.5$ exactly, any value in $[x, x_{\text{next}}]$ is a median; GCE usually expects the smallest integer x.)

6. Distribution of $X_1 + X_2$ (Convolution):

(a) Draw a two-way table of (x_1, x_2) with joint probabilities $p_1(x_1)p_2(x_2)$ (requires independence).

(b) Sum probabilities along diagonals where $x_1 + x_2 = z$.

(c) Present the resulting distribution of $Z = X_1 + X_2$.

7. Using the PGF:

(a) Write $G(t) = \sum p(x)t^x$.

(b) Differentiate: $G'(t) = \sum xp(x)t^{x-1}$, $G''(t) = \sum x(x-1)p(x)t^{x-2}$.

(c) Evaluate at $t = 1$: $E(X) = G'(1)$, $E(X^2) = G''(1) + G'(1)$.

(d) For sum of n i.i.d. RVs: $G_{S_n}(t) = [G_X(t)]^n$. Expand to read off the distribution if needed.

WARNING Common Mistakes & Traps

- Forgetting $\sum p(x) = 1$: Always verify or use it to find constants. Easy mark lost if omitted.
- Confusing PMF and CDF: $p(x)$ is point probability; $F(x)$ is cumulative. $P(X = x) = F(x) - F(x^-)$, not $F(x) - F(x-1)$ unless the support is integers with step 1.
- Variance Errors: $Var(aX) = a^2Var(X)$, not $aVar(X)$. $Var(X + Y) = Var(X) + Var(Y)$ only if X, Y are independent.
- Median Definition: For discrete RV, median is not always the "middle value". It is the smallest x with $F(x) \geq 0.5$. If $F(x) = 0.5$ exactly, any value in $[x, x_{next}]$ is a median; GCE usually expects the smallest integer x.
- Convolution Limits: When summing $P(X_1 + X_2 = z)$, ensure x runs over R_{X_1} such that $z - x \in R_{X_2}$. Do not assume infinite sums; the support is finite or countable.
- PGF Domain: $G(t)$ defined for $|t| \leq 1$. Differentiation valid inside radius of convergence. Always state $G(1) = 1$ as a check.
- Notation: Use X for RV, x for values. $E(X)$ is a number, not a function. Distinguish X_1, X_2 (different RVs) from x_1, x_2 (values).
- "Hence" in PGF Questions: If the question says "Hence find $E(X)$ and $Var(X)$" after finding $G(t)$, you must differentiate $G(t)$; computing directly from the table loses method marks.

KEY POINTS GCE Examination Tips

- Define the RV clearly: "Let X be the number of..." State support R_X explicitly.
- Show the $\sum p(x) = 1$ check: Even if obvious, write the sum and conclude = 1. Easy mark.
- Table Layout: Use a clear table with columns x, $p(x)$, $xp(x)$, $x^2p(x)$. Show a totals row. Neat tables gain method marks.
- Independence: Explicitly state "Since X_1, X_2 are independent..." before using

$p(x_1, x_2) = p_1(x_1)p_2(x_2)$ or $Var(X_1 + X_2) = Var(X_1) + Var(X_2)$.

- PGF Questions: Often ask to "Find $G(t)$", then "Hence find $E(X)$ and $Var(X)$". Do not compute $E(X)$ directly from table if "Hence" is used; must differentiate $G(t)$.
- Sum of n i.i.d.: Recognize $G_{S_n}(t) = [G_X(t)]^n$. If asked for distribution of sum, expand the polynomial/binomial to read off coefficients.
- Contextual Problems: Cameroonian contexts appear (e.g., "A bag contains 5 red and 3 blue balls…", "Number of customers at a Douala kiosk…"). Translate words into RV definition + distribution (Binomial, Geometric, Uniform, or ad-hoc).
- Working: Show all steps. $E(X^2) = \sum x^2 p(x)$ written out gets method marks even if arithmetic error later.
- Approximation: Not in this chapter (Poisson approximation to Binomial is in the next chapter). Stick to exact discrete calculations.

DID YOU KNOW? **Did You Know?**

The concept of a random variable was formalised by Andrey Kolmogorov in 1933 as part of his axiomatisation of probability theory. Before that, probabilities were tied to specific games or experiments. The Probability Generating Function was introduced by Pierre-Simon Laplace in 1812 for solving problems in combinatorics and probability; it is the discrete analogue of the Moment Generating Function used for continuous variables. In Cameroon, discrete random variables model many real situations: the number of successful candidates in a GCE session (Binomial), the number of taxi arrivals at a rank in 10 minutes (Poisson), or the number of draws until the first winning ticket in a loterie nationale (Geometric).

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